WEB-BASED SALES AND INVENTORY SYSTEM WITH MEMBERSHIP MANAGEMENT FOR BOSS RAP MOTOR SHOP

A Capstone Project Proposal
Presented to the Faculty of the
Information and Communications Technology Program
STI College Baliuag

In Partial Fulfilment
of the Requirements for the Degree
Bachelor of Science in Information Technology

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October 24, 2026

EXECUTIVE SUMMARY

This study focuses on developing a Web-Based Sales and Inventory System with Membership Management for Boss Rap Motor Shop. The main problem of the shop is that it still uses manual methods like notebooks to record sales and inventory. Because of this, errors happen, records can be lost, and transactions become slow. The proposed solution to address these issues is a web application that will enable the management of product inventories, transactions, and customer data in one database. The system will automatically update the inventory records, log sales transactions, and keep customer data. It will have a membership module, which will help identify regular customers and serve them better.

The purpose of the system is to enable efficiency and accuracy of the shop's operation. This will be useful to the shop owner in assessing the status of his business with regard to its performance. In the long run, the system will help satisfy the customers and ensure growth of the business. Overall, this study proves that computerizing a motor shop is more effective than manually running it.

ACKNOWLEDGEMENT

I would like to express my deepest gratitude to everyone who supported me in completing this thesis. First, I think God for giving me strength, wisdom, and guidance throughout this journey.

I am sincerely grateful to my adviser, who provided valuable advice, encouragement, and guidance during the development of this study. Their patience and support helped me improve my work and stay focused on my goals.

I also extend my thanks to my teachers and mentors who shared their knowledge and inspired me to continue learning. Their dedication has been a great influence on my academic growth.

To my family, I am forever thankful for their love, understanding, and support. They gave me motivation and strength to continue even during difficult times.

Lastly, I would like to thank my friends and classmates who offered help, encouragement, and companionship. Their support made this journey lighter and more meaningful.

This thesis would not have been possible without the guidance, support, and encouragement of all these people.

Approval Sheet

This thesis entitled “Web-based Sales and Inventory System with Membership Management for Boss Rap Motor Shop”, prepared and submitted by Mohammad Hamdie T. Langco, Vladimmhir B. Canchino, Aljae DJ. Cruz, Don Janken B. Teodoro in partial fulfillment of the requirements for the degree of Bachelor of Science in information Technology, has been examined and recommended for acceptance and approval

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Methodology 31
Research design 31
Research Locale 32
Respondents of the Study 32
Data Gathering Procedure 33
System Development Method 34
Calendar of Activities 36
Planning Phase 36
System Design Phase 36
Data Base Design 37
System Modules 37
Development Phase 41
Resources 43
System Architecture 44
Simple Use Case Diagram Structure 45
System Interface Design 45
Home Page 48
Shop Page 49
Service Page 50
Admin Dashboard Page 51
Inventory 52
Purchase Order 53
Admin Sales Pages 54
Sales Transaction Records 55
Current Order 56
Receipt for walk-in 57
Current Order Fegure 11 58
Method for Paying online 59
Fegure 13 60

Membership Management 61
Promotion and Discount Campaigns 62
Return and Replacement Request 63
Fegure 17 64
Customer Inquiries and Messages 65
User Account Management 66
Order Status Tracking 67
References 68
Curriculum Vitae 71

INTRODUCTION

Project Context

Motor shops play an important role in supporting motorcycle owners by providing spare parts and repair services. To operate efficiently and remain profitable, these businesses rely on effective management of their daily operations (Saputra & Fauzi, 2023). Among the most critical aspects of operations are inventory, sales, and customer management. According to Sudirga et al. (2021), inventory management refers to the process of monitoring product availability, stock levels, and the movement of items. Effective inventory management ensures that products are neither overstocked nor out of stock. Sales management, on the other hand, involves recording and tracking customer transactions, including product purchases, quantities, and total sales. Proper sales management ensures accurate financial records and smooth transaction processing (Ascon et al., 2023). Meanwhile, customer management focuses on handling customer information and interactions, which is essential for building relationships and encouraging repeat purchases (Aulia, 2022).

However, Conde et al. (2025) stated that many small-scale motor shops still rely on manual methods such as notebooks or basic spreadsheets to manage these operations. These methods are time-consuming and highly prone to human error. In terms of inventory management, manual recording may lead to inaccurate stock counts, making it difficult to determine product availability (Agboola et al., 2022). Similarly, a recent study indicates that manual sales management often leads to errors in recording transactions, which can result in incorrect sales data and financial inconsistencies (Yuniarto & Siregar, 2025). Furthermore, the absence of a structured customer management system makes it difficult to identify repeat customers or track their purchase history (Najwa et al., 2023). As a result, businesses are unable to implement effective strategies to retain customers or improve service quality. These issues can lead to delays in transactions, poor decision-making, and reduced customer satisfaction, ultimately affecting the overall profitability of the business.

In response to these challenges, many businesses are adopting computerized systems to

improve operational efficiency and data accuracy (Șișu et al., 2024). A web-based

system, in particular, provides a centralized platform where inventory, sales, and customer data can be stored, processed, and accessed in real time. By automating these processes, businesses can minimize errors, improve transaction speed, and enhance overall management capabilities (Divina et al., 2025).

The proposed project, titled Web-Based Inventory and Sales System with Membership Management for Boss Rap Motor Shop, aims to address these operational issues. The system will integrate three main components: inventory management, sales management, and membership management. The inventory management module will monitor stock levels and automatically update product quantities after each transaction. The sales management module will record and process customer purchases accurately and efficiently. The membership management module will store customer information, track purchase history, and identify repeat customers through a structured system. Additionally, the system will incorporate barcode scanning functionality to improve the speed and accuracy of product identification during inventory handling and sales transactions. It also supports quick customer identification for walk-in members.

Specifically, the membership component addresses the lack of a formal customer management approach in the shop. Currently, customer information is not systematically recorded, making it difficult to recognize loyal customers or analyze purchasing behavior. By introducing a membership system, the shop will be able to register customers, monitor their transaction history, and analyze purchasing behavior. This enables the implementation of reward mechanisms, identification of frequent customers, and development of strategies to increase customer retention.

Overall, the proposed system seeks to replace the existing manual processes of Boss Rap Motor Shop with an integrated web-based solution. By improving inventory accuracy, streamlining sales transactions, and establishing a structured membership system, the project aims to enhance inventory accuracy, accelerate transaction processing, and enable data-driven decision-making, ultimately enhancing customer satisfaction and business performance.

Purpose and Description of the Project

The primary purpose of this project is to design and develop a Web-Based Inventory, Sales, and Membership Management System for Boss Rap Motor Shop. The system aims to modernize and streamline the shop's daily operations by replacing its existing manual and semi-digital processes with a centralized, integrated, and computerized platform. Through this transformation, the project seeks to improve operational efficiency, enhance data accuracy, and support better decision-making for the business.

Based on the initial interview conducted with the shop owner, Boss Rap Motor Shop currently operates using a combination of manual recording and informal digital methods. Inventory and sales transactions are primarily tracked through handwritten records and basic monitoring practices. In addition, the shop actively accepts customer orders through its Facebook page, where customers inquire about product availability and place orders online. Once a purchase is confirmed, customers are given two options: they may either pick up the items at the store or request delivery, wherein the shop books a third-party courier service such as Lalamove, with the delivery cost shouldered by the customer. While this approach allows the business to reach a wider customer base, it lacks a structured system for recording transactions, tracking inventory changes, and managing customer data efficiently.

These existing practices present several operational challenges. Manual inventory tracking increases the risk of inaccurate stock counts, which may lead to overstocking or stock shortages. Similarly, sales transactions conducted through Facebook messaging are not automatically recorded in a centralized database, making it difficult to maintain consistent and reliable sales records. The absence of an integrated system also results in delays in retrieving information, potential duplication of data, and limited visibility of overall business performance. Furthermore, customer information from online transactions is not systematically stored, preventing the shop from identifying repeat customers or analyzing purchasing behavior.

In response to these challenges, the proposed system will provide a comprehensive and structured solution by integrating inventory management, sales processing, and membership management into a single web-based platform. The system will enable real-time monitoring of product stock levels, ensuring that inventory data is consistently

updated after every transaction. By automating inventory updates, the system reduces the likelihood of human error and provides accurate information on product availability at any given time.

For sales management, the system will support both walk-in and online transactions. It will allow staff to process in-store purchases efficiently while also recording orders that originate from the system. Each transaction will be automatically stored in the system, ensuring that all sales data is consolidated in one database. This improves the accuracy of financial records and enables faster transaction processing. Additionally, the integration of barcode scanning functionality will further enhance operational efficiency by allowing quick product identification during sales and inventory handling, reducing reliance on manual input

The system will also incorporate a membership management feature to address the lack of structured customer data handling. Through this feature, customers can be registered within the system, and their information, including contact details and transaction history, will be securely stored. This allows the business to identify repeat customers, monitor purchasing patterns, and implement strategies aimed at improving customer retention. Membership records may also support features such as customer identification during transactions and the potential implementation of rewards or promotional programs, thereby enhancing customer engagement and loyalty.

Another important component of the system is its reporting capability. The proposed platform will generate comprehensive and organized reports on sales performance, inventory status, and product movement. These reports will provide valuable insights that can assist the shop owner in evaluating business performance, identifying fast-moving or slow-moving products, and making informed decisions regarding stock replenishment and sales strategies. By having access to accurate and timely reports, the business can improve its planning and overall management.

Moreover, the web-based nature of the system ensures accessibility and flexibility. Authorized users can access the system through an internet connection, allowing the shop owner to monitor operations even outside the physical store. This is particularly beneficial given that a portion of the shop's transactions already occurs online through Facebook, as it creates a more seamless integration between online inquiries and recorded sales data.

Overall, the proposed Web-Based Inventory, Sales, and Membership Management System aims to replace the fragmented and manual processes currently used by Boss Rap Motor Shop with a unified and efficient solution. By integrating key business functions into a single platform, the system is expected to improve data accuracy, reduce operational delays, enhance customer management, and support data-driven decision-making. Ultimately, the project seeks to contribute to the growth and sustainability of the business by providing a reliable technological foundation for its operations.

Objective of the Study

General Objective

The main objective of this study is to design and develop a web-based inventory, sales, and membership management system that enhances the efficiency, accuracy, and reliability of product, transaction, and customer data management for Boss Rap Motor Shop. The system aims to address the limitations of the shop's current manual and informal processes by providing an integrated platform that supports both in-store and online operations.

Specific Objectives

To achieve the general objective, the study aims to accomplish the following:

To develop a comprehensive inventory management module that ensures accurate and real-time monitoring of product stocks.
This module will enable the efficient management of product records by allowing authorized users to add, update, and organize product information. It supports accurate monitoring of stock levels and ensures that inventory data is consistently maintained. The system aims to improve inventory control, reduce errors, and enhance the overall efficiency of stock management within the business.

To develop an integrated sales management module that processes and records transactions efficiently and accurately.
This module will facilitate the efficient processing and recording of sales transactions. It ensures that transaction details are properly captured and organized, supporting accurate sales monitoring and record-keeping. The system aims to improve transaction efficiency, reduce errors, and provide reliable sales data for business operations.

To develop a membership management module with QR code-based identification to support structured customer data handling and customer retention.

This module will enable the registration and secure storage of customer information, including names, contact details, and transaction history. It will support the identification of repeat and high-value customers and facilitate the analysis of purchasing behavior. The inclusion of QR code-based identification will enhance customer recognition and improve transaction efficiency, contributing to better customer engagement and retention.

To develop a reporting module that provides accurate and comprehensive business performance insights.

This module will generate reports on sales performance, inventory status, and customer activity. It will assist the shop owner in monitoring product movement, identifying fast-moving and slow-moving items, analyzing sales trends, and supporting data-driven decision-making for business improvement.

To develop an integrated system that connects the sales, inventory, purchase order, and reporting modules

This module ensure accurate and synchronized financial and transaction data within the system. The purchase order process allows the admin to create orders for suppliers, track item deliveries, and automatically update the inventory and reports when products are received.

Scope and Limitation of the Study

Scope

This study focuses on the development of a Web-Based Inventory, Sales, and Membership Management System for Boss Rap Motor Shop. The system will cover the integration of inventory management, sales processing, membership management, reporting, and digital payment recording into a centralized platform to improve operational efficiency and data accuracy.

- **User Authentication and Access Control Module**

This module allows authorized users such as the shop owner and employees to register and securely access the system through a login process. It includes role-based access control to ensure that only permitted users can perform specific system functions.

- **Inventory Management Module**

This module allows the management of product records, including adding, updating, and categorizing products. It supports stock-in and stock-out processes and automatically updates inventory levels after each transaction. The system also integrates barcode scanning for product identification, enabling faster processing of inventory updates and sales transactions.

- **Sales Management Module**

This module handles both walk-in and online transactions, including orders received through Facebook inquiries. It records product purchases, quantities, prices, and transaction details. The system automatically computes total sales, generates transaction records, and updates inventory in real time. Barcode scanning is used during walk-in transactions to improve speed and accuracy in product selection and processing.

- **Membership Management and Loyalty Module**

This module allows the registration and secure storage of customer information, including names, contact details, and transaction history. It supports customer tracking and the identification of repeat customers. The system generates QR code-based membership cards that are used for customer identification during transactions. The module also supports basic membership benefits such as loyalty tracking and customer engagement features to encourage repeat purchases.

- **Reporting Module**

This module generates comprehensive reports on sales performance, inventory status, product movement, and customer activity. It provides insights such as fast-moving and slow-moving items and supports data-driven decision-making for business operations.

- **Online Payment Recording Module**

This module records and tracks digital payments made through e-wallets and online banking platforms. It ensures that all online transactions are properly documented and linked to corresponding sales records for accurate financial tracking and reconciliation.

- **Purchase Order**

This module records and tracks purchase orders made to suppliers for restocking products and supplies. It ensures that all ordered and received items are properly documented and automatically connected to the inventory and reporting records for accurate tracking.

- **Inventory Management**

  The system allows the administrator to maintain a digital product catalog, track stock levels in real-time, scan items using Barcode/QR code integration, and manage supplier information.

- **Membership and Loyalty Program**

  To enhance customer retention, the system features a customer profiling tool with a tiered membership structure (**Regular, VIP, and Premium**). It automates loyalty point accumulation, applies member-exclusive discounts at checkout, and handles membership renewals.

- **Point of Sale (POS)**

  The POS module processes sales transactions via multiple payment methods, including Cash and E-wallets (e.g., GCash, Maya). It features a quick check-out interface and automatically generates digital or printed invoices.

- **Service Management**

  or workshop operations, the system tracks historical vehicle service records, creates digital work orders for mechanics, schedules customer appointments, and computes labor costs based on technician hours.

- **Reporting and Analytics**

  The system provides management with data-driven insights through automated reports, including Sales performance, Inventory movement, Customer insights (VIP preferences), and Profit/Loss statements.

- **Notification System**

  Automated alerts are integrated to notify administrators of low stock levels, remind members of upcoming membership expirations, and blast promotional announcements.

- **Return and Replacement**

  The system includes a dedicated workflow to process item returns and replacements in accordance with shop policies.

- **Printable Inventory Stock**

  The system includes a printable inventory stock sheet to record and track item quantities, stock in, stock out, and balances for easy stock management.

Limitations

The study has the following limitations:

- **Maintenance and System Updates**

The system requires regular maintenance and updates to ensure continued security, data integrity, and optimal performance. These updates may include system enhancements, bug fixes, and database optimization. However, the maintenance and administration of the system will be the responsibility of the system administrator or designated authorized personnel of Boss Rap Motor Shop after deployment. The proponents will no longer handle post-deployment maintenance once the system is implemented and turned over to the client.

- **Internet Dependency (Online System Requirement)**

The system is web-based and requires a stable internet connection to function. In cases of internet disruption or power outages affecting connectivity, real-time system access and transaction recording will be temporarily unavailable. During such instances, transactions may be recorded manually and encoded into the system once connectivity is restored.

- **Initial Manual Data Encoding for Product and Customer Information**

The system requires initial manual encoding of product details and customer information during setup and registration. While barcode scanning is used during transactions, product data must still be manually encoded into the system beforehand.

- **Limited Automation in Membership Perks System**

Although the system includes a QR code-based membership module, the loyalty or reward system is limited to basic tracking of customer purchases and does not include advanced automated reward computation or complex tier-based membership levels.

For example, the system does not support advanced reward mechanisms such as progressive point multipliers (e.g., increased points for VIP or high-value customers), dynamic discount tiers based on cumulative or monthly spending thresholds, or automated promotional rules such as “buy X get Y” schemes triggered by customer purchase history or behavioral analysis.

- **Dependence on User Input Accuracy**

The accuracy of inventory, sales, and customer records depends on correct user input. Any incorrect encoding of data may affect reports, inventory counts, and transaction records.

REVIEW OF RELATED LITERATURE/STUDIES/SYSTEMS

Web-Based Inventory System Design at Toko Sahabat Using RAD Method

The study which was titled "Web-Based Inventory System Design at Toko Sahabat Using RAD Method," discussed how a small retail store was able to develop a web-based inventory system that would improve the store's day-to-day operations. According to the researchers, the store was manually recording the inventory. However, the researchers found that the manual recording resulted in errors in the counting of the store's inventory. It also resulted in delays in the checking of the products' availability. Because of the problems that the store encountered with the manual recording, the researchers found it worth creating a web-based inventory system. The web-based inventory system was created through the Rapid Application Development, or RAD, method. This method aims to develop systems rapidly. The system allows the real-time monitoring of the inventory. The store owner can monitor the products at any given time. The system reduces the chances of error in recording. The system also increases the speed of transaction. The reports can be generated automatically. These reports help the owner understand the business. The system can be accessed through the browser. The system is not complex in terms of installation. The study found that the web-based inventory system increased efficiency. The workload is reduced for the employees. The accuracy is improved. The study proves the usefulness of computerized systems for small businesses.

This study is similar to the proposed system for Web-Based sales and inventory system with Membership Management for Boss Rap Motor Shop because both systems focus on web-based inventory management. Both systems aim to replace manual recording methods. Both reduce human errors in stock monitoring. Both allow adding, editing, and deleting product information. Both automatically update stock levels after sales. However, the difference is that the Toko Sahabat system mainly focuses on inventory management only. The proposed system for Boss Rap Motor Shop includes inventory, sales, and membership management. Another difference is that the proposed system includes a membership card feature.

The Toko Sahabat system does not have a customer membership management function. A sales transaction module is also part of the proposed system. The study proves the necessity of automating a small business. It proves the efficiency of the web-based system. It also proves the significance of real-time monitoring. These ideas were incorporated into the proposed system. It enhances the management of the stock in a motor shop business. It monitors the motorcycle parts and accessories. It also enables faster transaction. The study provides a solid base for the proposed research. It proves the efficiency of the web-based system. It proves the significance of the proposed system. It proves the efficiency of the proposed system.

Web-Based Inventory and Sales Information System: MSME Case Study

The study under the title "Web-Based Inventory and Sales Information System: MSME Case Study" discussed the development of a web-based information system for small and medium enterprises. The study revealed that many small and medium enterprises were using manual systems. Manual systems were causing wrong sales recording. Manual systems were also causing wrong inventory recording. Due to this problem, the researchers developed a web-based inventory system. It records sales transactions. It also updates inventory after each sale. It records all sales transactions in a database. It allows the business owner to generate reports. These reports include daily sales, weekly sales, and monthly sales. It makes the calculations accurate. It saves time. It requires an internet connection to access the system. It makes the business records transparent. The study revealed that the system increased productivity. It reduced the loss of data. It improved the business decision. The business owner can see the sales trend. It increased the efficiency of customer service. It improved the overall business operation. It integrated the inventory system with the sales system. The system helps improve accuracy in the calculations. It helps reduce the time taken for manual work. The system can be operated through an internet connection. It helps improve transparency in business transactions. The researchers established that the system improved productivity. It reduced data loss. It improved decision-making. Business owners can now easily identify sales trends. The system helps improve efficiency in customer services. The study indicates that a combined inventory and sales system helps improve small business operations.

Web-Based Stock Management and Sales Information System for Grocery Store

The study, Web-Based Stock Management and Sales Information System for Grocery Store, demonstrated the ways in which a grocery store was able to improve its operations through a computerized system. The researchers noted that the grocery store used notebooks to record sales and inventory levels. The use of notebooks resulted in human error in counting the stock levels. It also resulted in a delay in serving the customers. The researchers, therefore, decided to design a computerized system to help the grocery store improve its operations. The new system helps in managing the stock levels and sales records of the grocery store. It updates the stock levels after each sale. The new system helps in storing the information of the products in a database. It helps in quickly searching the products using the database. It also helps in generating reports of sales. It improves the speed of transactions. The store owner can monitor business performance anytime. The system can be accessed through a browser with internet connection. It improves accuracy in stock monitoring. It also prevents overstocking and shortages. The study found that employees became more productive. Customers experienced faster service. The system reduced paper usage. It also secured business data. Overall, the study concluded that a web-based system improves retail store management.

This study is similar to the proposed system for Boss Rap Motor Shop because both focus on stock management and sales recording. Both systems replace manual recording with a computerized system. Both automatically update inventory after each transaction. Both generate sales reports. However, the grocery store system focuses only on stock and sales management. The proposed system for Boss Rap Motor Shop includes membership management and membership card generation. Another difference is that the grocery system is designed for food items, while the proposed system is for motorcycle parts and accessories. The proposed system also tracks loyal customers. The study supports the importance of automation in small retail stores. It shows that real-time stock monitoring reduces errors. The proposed system applies the same concept. It improves inventory tracking in a motor shop. It ensures accurate sales recording. It also enhances customer service. The study strengthens the idea that web-

based systems increase efficiency.

Web-Based Sales and Inventory System for a Stationary Store

In the study "Web-Based Sales and Inventory System for a Stationery Store," the authors talked about creating a system where sales and inventory management can be done in one platform. The researchers found out that the stationery store was having problems in monitoring the items in stock. This was caused by manual recording of the items. As a result, some information was lost, and the information was inaccurate. Hence, the researchers created a system where sales and inventory can be done online. The system can record all sales transactions. The system can calculate the totals automatically. The system can update the stock levels in real-time. The system can add and modify products. The system can store information in a safe database. Reports can be generated at any time. The system speeds up the transaction. It eliminates human computation errors. The store owner is able to access reports remotely. The system improves the overall business monitoring. The employees do not need to calculate the totals. The customer service improved according to the findings. The system eliminates data loss. It also improves the overall efficiency. In summary, the system was effective for the retail business management.

The similarity of this study to the proposed system is that this study and the proposed system integrate the sales and inventory management module. This study and the proposed system are web-based systems. This study and the proposed system are used to improve accuracy and efficiency. However, this study does not integrate membership management. In the proposed system, there is membership registration and membership cards. Another difference is that this study deals with stationery store products, while the proposed system deals with motorcycle products. In the proposed system, there is tracking of loyal customers. This study supports the use of automation in retail stores. This study confirms that real-time updates are essential. In the proposed system, this principle is applied. In the proposed system, stock is updated automatically after every transaction. In the proposed system, there is a report generated to monitor transactions. In the proposed system, transactions are fast. This study supports the need to use digital systems in small businesses. This study supports the proposed research.

Research on the Application of Membership System in Market Management

The study, which is titled "Research on the Application of Membership System in Market Management," indicates the role of membership systems in promoting customer loyalty in businesses. The researchers used various retail markets that use membership systems in their research. The researchers noted that membership systems help in customer retention. The customer feels appreciated when they join the membership program. The membership program helps in recording customer information and tracking their purchases. It helps in offering discounts to customers. The membership program helps in understanding customer behavior. The business can identify their frequent customers and reward loyal customers. The researchers noted that membership systems increase sales, customer satisfaction, and enhance marketing strategies. The membership program helps in storing data in a database, creating a membership report, and evaluating business performance using the data. The researchers concluded that membership systems are beneficial to retail businesses.

This study is closely related to the proposed system for Boss Rap Motor Shop because it includes a membership module. Both systems store customer information. Both track purchase history. However, the study focuses mainly on membership marketing strategies. The proposed system focuses on inventory, sales, and membership integration. The proposed system also generates membership cards. Another difference is that the study discusses general retail markets, while the proposed system is for a motor shop. The study supports the importance of customer loyalty programs. It confirms that membership systems increase customer satisfaction. The proposed system applies this concept. It helps identify loyal customers. It improves customer relationships. It also supports promotions in the future. The study strengthens the need for membership management. Therefore, it supports the proposed system development.

Review of Related Studies

Inventory Management System for Motorcycle Dealers

The study, titled “Inventory Management System for Motorcycle Dealers,” deals with the inventory management of spare parts in a motorcycle dealer’s store by using a computerized system. According to the researchers, inventory management is an integral part of a motorcycle dealer’s business. It is important for the dealer to keep an eye on the stock of spare parts. This is because, without proper inventory management, the dealer might face losses. Some motorcycle dealers are having difficulties in managing their spare parts. Because of this, the researchers have come up with an inventory management system. This system includes the entire process of the product, from purchase to sale. It records the movements of the product in real time. The researchers identified the needs of the business before designing the system. A model for tracking the inventory data was also created. The system includes modules for stock monitoring and inventory control. It also facilitates the management of warehouse activities. It enables businesses to be aware of when they need to purchase more stock. It also monitors returned stock and new stock arriving. The researchers tested the system. This ensured that it worked as it should. It showed that the system improved the accuracy of the inventory. It also helped businesses monitor the products more effectively. It was concluded that inventory management systems are important for motorcycle dealers.

This study is related to the proposed system for Boss Rap Motor Shop in the following ways: First, both systems deal with the management of motorcycle spare parts. Second, the systems are designed to enhance the management of spare parts. Third, the systems are based on computerized systems rather than manual recording. Fourth, the systems deal with the flow of products from purchase to sale. However, there are differences between the proposed system and the study. First, the study is based on inventory management for motorcycle dealers. Second, there are no sales transaction modules and membership in the study. Third, the proposed system deals with inventory, sales, and membership. Fourth, there are differences in terms of the business environment. In general, motorcycle dealerships deal with selling motorcycles and other motorcycle

parts in larger

setups. However, the focus of the proposed system is on the local business of a motor shop. Yet, despite these differences, this study offers full support for the proposed research project. First, it affirms that proper management of inventories is vital for motorcycle businesses. Second, it affirms that computerization enhances accuracy and efficiency in any business process. These will be the same concepts that will be applied in the proposed system for Boss Rap Motor Shop. The proposed system will aid in better management of motorcycle parts inventories. It will also aid in better sales recording and customer management.

Design of wed-based spare parts inventory information system using the rad method

The study, titled "Design of Web-Based Spare Parts Inventory Information System Using the RAD Method," discussed how an inventory and sales system was developed for an automotive company that deals with spare parts. The study was based on the observations of the researchers that the company was using a manual recording system for their inventory of spare parts. Under this system, employees of the company had to record their daily and monthly inventory reports. Due to this, employees faced problems in recording their reports. They also faced problems in searching for product-related information, as this required a lot of time. Thus, the study observed that due to this manual recording, efficiency was compromised. There was also a higher probability of human error. To overcome these problems, a web-based inventory information system was designed. Under this system, users can easily store their spare parts-related data. They can also record their inventory automatically. The system has been developed using the RAD (Rapid Application Development) method. The RAD method of developing a system emphasizes the rapid creation and testing of the system. The system has been provided with user access control, which adds security to the system. It is easy for the managers to generate reports using the system. These reports help the managers to keep track of the inventory and the movement of products. The system has been able to increase the speed of retrieving information. It is easy for the employees to search for the information regarding the spare parts. The researchers found that the system increased efficiency and accuracy. Additionally, the system reduced the workload of employees. On the whole, the study concluded that a web-based spare parts inventory system is beneficial to the operations of automotive businesses.

This study is related to the proposed system, namely, Web-Based Inventory, Sales, and Membership Card System for Boss Rap Motor Shop, in that both systems are centered on managing the inventory of spare parts of a business establishment through a web-based system. Both systems are intended to replace manual recording with a computerized system. Both systems utilize a digital database to store product-related data. Another similarity between the two systems is that both are intended to increase efficiency and reduce human error. However, there are also some differences between

the two systems. One of these differences is that, in the system presented in this study,

the main features are centered on managing the inventory of spare parts and sales reports. There is no feature of membership. In the proposed system for Boss Rap Motor Shop, there are features of membership registration and generation of membership cards. Another difference is the kind of business environment. The study is based on an automotive spare parts company, including warehouse operations. The proposed system, on the other hand, is based on a local motor shop that sells motorcycle parts directly to customers. Despite the differences, the study still supports the proposed research. This is because it confirms that the proposed system, which is web-based, will improve the management of the company's inventory. Moreover, it confirms that the proposed system, which is web-based, will enable the company's employees to work faster and more accurately. These are the same principles that the proposed system will apply. It will help Boss Rap Motor Shop manage its inventory, sales, and customers' membership. For this reason, the study supports the proposed research.

Web and mobile-based motorcycle spare part sales system (case study: rey motor workshop)

The study, which is titled “Web and Mobile-Based Motorcycle Spare Part Sales System: Case Study of Rey Motor Workshop,” aims to create a digital system for the motorcycle workshop. The researchers observed that the motorcycle workshop still employed traditional manual methods. The customers had to come to the shop physically to check the available spare parts. However, the process sometimes became inconvenient for the customers. Some customers even had to come to the shop only to find out that the spare parts they needed were not available. The motorcycle workshop also had difficulty in monitoring the inventory and the transactions. Due to the problems encountered, the researchers were able to create a web and mobile-based system for the motorcycle workshop. The system allows the recording of transactions electronically. It also allows the monitoring of the inventory of the spare parts through the database. This will enable customers to know the available parts. The system will also enhance communication between the workshop and customers. The researchers used the Waterfall method to design the system. The system will have different modules for managing products and recording transactions. The study found that the system enhanced the workshop's efficiency. It reduced the loss of data. It also enhanced customer services. The study concluded that digital systems enhance the operations of motorcycle workshops.

This study is related to the proposed system for Boss Rap Motor Shop in that both systems are concerned with improving the management of motorcycle spare parts and sales transactions. Both systems utilize technology to replace manual recording. Both systems also utilize a database to monitor their respective records. Both systems are also concerned with improving efficiency and serving customers better. However, there are also similarities between the two systems. One of these similarities is that the system in the study is concerned with the sales of motorcycle spare parts. There are no membership features in this system. The system that was proposed for Boss Rap Motor Shop includes a membership system and generation of membership cards. Another difference that was noticed was that, whereas the study was based on both web and mobile platforms, the proposed system was based mainly on a web platform. Despite

these differences, the study supports the proposed research. It shows that technology

can improve the workings of motorcycle workshops. It also shows that an automated inventory control and sales system can improve efficiency. These are also the concepts that are going to be implemented in the proposed system. They are going to improve the workings of Boss Rap Motor Shop.

Development of Spare Part Management System for Automotive Repairing Shop

The research paper entitled “Development of Spare Part Management System for Automotive Repairing Shop” aims to create a computerized system for the management of the inventory of the automotive repairing shop’s spare parts. The researchers have observed that the automotive repairing shops have some difficulties in managing the inventory of the spare parts and the sales information. The recording of the information manually often creates a problem in the clarity of the information about the products. The staff also find it difficult to keep track of the information manually. Therefore, the researchers have developed a system for the management of the spare parts inventory. The developed system includes the features for the stock monitoring and the sales information. This makes the system more secure. The system also has a search facility, which enables users to easily find the spare parts they need. The researchers aimed to create a system that would improve the management of inventory and the recording of transactions. The system also helps in the systematic management of the information regarding the spare parts. It helps to reduce the use of paper. It also improves the working conditions for the employees. The study showed that the system improves the efficiency of the employees and reduces the amount of work they do. The employees were able to easily manage the information regarding the spare parts. The study concluded that computerized systems for the management of spare parts improve the operations of automotive repair shops.

This study is somehow related to the proposed system for Boss Rap Motor Shop because the two systems share a common purpose, which is the management of the inventory of the spare parts using a computerized system. Both systems aim to reduce the recording and increase the efficiency in the management of the inventory. Both systems also use a database in the storage of the products and the transaction records. Another similarity between the two systems is the inclusion of the stock and sales management in the proposed system. However, there are differences between the two systems. The study does not include the membership management feature. The proposed system will have features such as membership management and issuance of membership cards. Another difference is that the study is based on general automotive repair shops. However, the proposed system is based on a motorcycle spare parts shop.

With these differences, the study still supports the proposed research. This is because

it affirms the importance of computerized systems in the management of inventories. It also affirms the importance of computerized systems in enhancing efficiency in automotive businesses.

WEB Based Motorcycle Sales Information System

The study on the information system titled “Web-Based Motorcycle Sales Information System” aims to enhance sales management in a motorcycle business. The researchers have seen that the company is using a spreadsheet in recording sales information. This method is hard for employees in searching the information. It also takes a long time in printing sales reports. Because of these problems, the researchers proposed a web-based information system. This information system records sales transactions of motorcycles. It also stores sales information in a database. This information system can produce reports. These reports help in monitoring business performance. It also enhances the speed in searching information. Employees can easily access sales information. The system helps improve the organization of sales records. It reduces the loss of important sales records. The researchers observed that the system improved the efficiency of sales management. Employees found it easy to manage sales records. The system improved the efficiency of report generation. In general, the study found that web-based sales systems are beneficial for the management of motorcycle sales.

This study is related to the proposed system for Boss Rap Motor Shop because both systems have similarities in terms of handling motorcycle sales using a web-based system. Both systems are designed to replace manual recording systems with digital systems. Both systems are also designed to produce reports that will help business owners monitor sales performance. However, there are differences between the proposed systems. The focus of the system in the study is mainly on motorcycle sales transactions. The focus of the proposed system is different from that of the study in that it includes inventory management and membership cards. The focus of the study is also different from that of the proposed system in that it does not have inventory management or membership cards. However, it is clear that the study supports the proposed research in that it indicates that web-based systems enhance business efficiency.

Synthesis

Based on the related literature and related studies reviewed, it is obvious that many businesses nowadays are using computerized systems to improve their daily operations, The studies explained that the manual system of recording data is a problem in the management of a business. The problems include errors in the inventory list, slow processing of transactions, and difficulty in organizing the data of the business, Because of this problem, many researchers have developed a web-based inventory system to improve the operations of the business, The system will be able to store the data of the products, sales, and customers, This will help the business owner to monitor the business more effectively, The system will also be able to access the data anywhere through the internet, The studies have shown that the computerized system will reduce errors in the business, The system will also be able to improve the productivity of the business. The reports will be generated through the computerized system, The reports will be used to improve the decisions of the business owner.

The proposed system for Boss Rap Motor Shop is similar to the systems discussed in the literature and studies because it also uses a web-based platform to manage business operations, Like the systems in the reviewed studies, the proposed system will manage inventory and record sales transactions. It will also generate reports to help the shop owner monitor business performance. However, the proposed system is different because it also includes a membership management feature and membership card generation, This feature allows the shop to store customer information and identify loyal customers, The membership card can also help the shop improve customer relationships and provide better service, Therefore, based on the reviewed literature and studies, the development of the Web-Based Inventory, Sales, and Membership Card System for Boss Rap Motor Shop is supported, The reviewed literature and studies have proven that web-based systems can efficiently improve the efficiency of a business by reducing manual work and improving the organization of business data, The reviewed literature and studies have also proven that a computerized system can efficiently improve the operation of a business, thus the importance of the proposed system for the operation of the business.

Table of Comparison

Criteria	Proposed System: Web-Based Inventory, Sales and Membership Card System for Boss Rap Motor Shop	Sari et al. (2020)	Putra et al. (2021)	Rahman et al. (2021)	Ahmad et al. (2020)	Yusuf et al. (2022)
Web-Based System?	Yes	Yes	Yes	Yes	Yes	Yes
Focuses on Sales Analytics?	Yes	No	Yes	Yes	No	Yes
Includes E-Payment?	Yes	No	No	No	No	No
Improves Inventory Management?	Yes	Yes	Yes	Yes	Yes	No
Enhances Customer Experience?	Yes	Yes	Yes	Yes	No	Yes
Helps Enhance Productivity?	Yes	Yes	Yes	Yes	Yes	Yes
Addresses Security Concerns?	Yes	Yes	Yes	Yes	Yes	Yes
Has Membership Management?	Yes	No	No	No	No	No

METHODOLOGY

Research Design

This study employs a descriptive developmental research design. The descriptive method is used to examine and understand the existing processes of Boss Rap Motor Shop in managing inventory, sales transactions, and customer information. It focuses on analyzing the current manual and semi-digital practices, particularly in recording product stocks, processing sales, and handling customer data.

On the other hand, the developmental method is used in the design and development of a Web-Based Inventory, Sales, and Membership Management System for Boss Rap Motor Shop. This method focuses on creating a functional system that addresses the identified issues in the current operational setup and improves business processes through system automation and integration.

The purpose of this research design is to assess the current workflow of the shop and serve as a basis for developing a system that improves efficiency, accuracy, and reliability in business operations. Data were gathered through interviews, observation, and document review to identify issues such as manual encoding errors, fragmented records, and difficulty in tracking transactions and customer history.

Overall, this research design ensures that the proposed system is grounded on actual operational needs and is capable of improving the overall management of inventory, sales, and customer information at Boss Rap Motor Shop.

Research Locale

The study was conducted at Boss Rap Motor Shop, a local business that sells motorcycle parts and accessories. The shop currently operates using manual recording methods for inventory and sales transactions, supplemented by informal digital communication such as Facebook inquiries for customer orders.

At present, product stocks and sales are recorded through handwritten logs, which limits real-time monitoring and increases the possibility of errors and inconsistencies. Customer information from both in-store and online transactions is also not systematically stored, making it difficult to track purchase history and repeat customers. These limitations highlight the need for a more organized and centralized system for managing business operations. In response, the proposed Web-Based Inventory, Sales, and Membership Management System aim to improve data handling by providing a centralized platform for inventory, sales, and customer membership records, thereby enhancing operational efficiency and accuracy at Boss Rap Motor Shop.

Respondents of the Study

The respondents of the study were the shop owner and selected employees of Boss Rap Motor Shop. These individuals were chosen because they are directly involved in the daily operations of the business, particularly in inventory management, sales processing, and customer handling.

The study used purposive sampling, selecting respondents who have direct knowledge and experience in the shop's operational processes. Their participation provided essential information on how inventory is recorded, how sales transactions are processed (including walk-in and Facebook-based orders), and how customer data is currently managed.

Their responses were crucial in identifying existing challenges in the current system, such as difficulties in monitoring stock levels, delays in recording transactions, and the lack of a centralized customer management system. These findings served as the basis for developing the proposed Web-Based Inventory, Sales, and Membership Management System for Boss Rap Motor Shop.

Data Gathering Procedure

The researchers employed several methods to collect the necessary data for the study. These methods were used to gain a comprehensive understanding of the current operational processes of Boss Rap Motor Shop, particularly in terms of inventory management, sales transactions, and customer information handling.

- **Interview**

The researchers conducted interviews with the shop owner and selected employees of Boss Rap Motor Shop. The interview aimed to gather detailed information regarding the current processes used in managing inventory, recording sales transactions (including walk-in and Facebook-based orders), and handling customer information. It also helped identify the challenges and problems encountered in the existing manual system, such as difficulties in tracking stock levels, delays in transaction recording, and lack of centralized customer data.

- **Observation**

The researchers observed the actual day-to-day operations of the shop, focusing on how inventory is recorded, how sales transactions are processed, and how customer inquiries are handled. This method allowed the researchers to directly understand the workflow of the shop, including manual encoding practices and the handling of both in-store and online transactions. Observation provided firsthand insight into inefficiencies in the current system that are not always fully expressed during interviews.

- **Document Review**

The researchers reviewed existing documents such as product lists, handwritten inventory records, and sales logs used by the shop. These documents were analyzed to understand how data is currently stored, organized, and updated. The review helped identify inconsistencies, redundancy, and limitations in the manual recording system, which supported the justification for developing a centralized web-based system.

System Development Method

The researchers used the **Waterfall Model** in developing the proposed system. The Waterfall Model is a linear and sequential software development approach in which each phase must be completed before proceeding to the next. This model is suitable for the study because it follows a structured process in developing a Web-Based Inventory, Sales, and Membership Management System for Boss Rap Motor Shop.

- **Requirement Analysis**

In this phase, the researchers gathered information about the current system used by Boss Rap Motor Shop. Data were collected through interviews, observation, and document review to identify the problems encountered in the manual and semi-digital processes. The researchers analyzed issues related to inventory management, sales recording (including Facebook-based transactions), and customer information handling. The system requirements were then determined based on these findings, including the need for inventory automation, barcode-based product identification, QR code-based membership, and reporting features.

- **System Design**

In this phase, the researchers designed the overall structure of the system. This includes the design of the database structure, user interface, and system modules such as inventory management, sales management, membership management, reporting, and online payment recording. The design also defined how barcode scanning will be used for product identification and how QR codes will be implemented for customer membership cards.

- **System Development**

In this phase, the actual development of the web-based system was carried out. The researchers used appropriate programming languages and database tools to build the system according to the approved design. The modules for inventory, sales, membership, reporting, and payment recording were developed and integrated into a centralized platform.

- **System Testing**

After development, the system was tested to ensure that all functionalities work correctly. This includes testing of inventory updates, sales transactions, barcode scanning for product identification, QR code scanning for membership verification, and report generation. Errors and issues identified during testing were corrected to improve system performance, accuracy, and reliability.

- **System Implementation**

After successful testing, the system was implemented and deployed for evaluation at Boss Rap Motor Shop. The system was presented to the shop owner and end users for initial use and assessment. Feedback from the implementation stage was considered for further improvement and refinement of the system.

Technical Background

Overview of Current Technologies to be Used in the System

At present, the daily business operations of Boss Rap Motor Shop continue to rely heavily on manual and informal recording practices. Inventory quantities, stock movements, and item sales are documented through handwritten physical logbooks and paper ledgers. When customers inquire about spare parts availability or pricing, staff must manually inspect storage shelves or review physical lists, leading to transaction bottlenecks and occasional stock discrepancies. Furthermore, although the shop actively accommodates customer inquiries and parts orders through its official Facebook page, these transactions are recorded manually and are not connected to a unified database. Customer contact details, transaction histories, and repeat purchase patterns are not systematically tracked, leaving the business without a structured customer relationship and loyalty management mechanism. Manual calculations also increase the risk of computation errors, while sales summaries require laborious manual compilation at the end of each business day.

To address these operational constraints, the proposed Web-Based Sales and Inventory System with Membership Management introduces modern, scalable web and database technologies. The system's frontend is constructed using React 19, TypeScript, and Vite, delivering a highly responsive single-page application (SPA) optimized for fast page loads and smooth user interactions across counter desktop terminals, tablets, and mobile smartphones. User interface navigation is structured through React Router DOM with role-based access control guards that restrict administrative functions to authorized personnel, while providing dedicated views for staff cashiers, registered members, and logistics personnel.

On the backend and data storage layer, the platform integrates with Supabase, leveraging a managed PostgreSQL 15 relational database. This architecture provides enterprise-grade ACID transaction guarantees, strict foreign key referential integrity across sales and inventory records, and secure cloud data hosting. Real-time database updates ensure that product stock levels are automatically decremented upon checkout completion at the Point of Sale (POS) counter. Additionally, Supabase Storage buckets are utilized to securely store uploaded payment receipts, item photos, and return claim

proofs. To enhance in-store counter speed, the system integrates hardware barcode scanning emulation, allowing cashiers to scan standard EAN-13 barcodes directly into the POS input or execute quantity multipliers with standard scanner keyboards. For customer membership, the system generates dynamic, unique QR code cards that cashiers scan to instantly retrieve customer loyalty profiles and apply tiered membership perks.

Complementing these transactional modules is an automated SMS notification gateway powered by the Semaphore SMS API service (under sender name TranslertPH). This service dispatches immediate transactional SMS alerts to Philippine mobile numbers whenever an online order is placed, when payment is verified, when parcels are marked 'Out for Delivery' with designated rider information, and upon final delivery drop-off. By combining modern web frontend engineering, relational cloud database management, barcode/QR hardware integration, and automated telecommunication alerts, the system replaces vulnerable manual paperwork with a robust, transparent, and scalable operational platform for Boss Rap Motor Shop.

Calendar of Activities Planning Phase

In this phase, the researchers analyzed the current system used by Boss Rap Motor Shop. The study revealed that the shop relies on manual recording for inventory and sales transactions, as well as informal handling of customer orders through Facebook inquiries. This process often results in data inconsistencies, delays in transaction recording, and difficulty in monitoring product availability and sales performance.

Based on the identified problems, the researchers planned the development of a Web-Based Inventory, Sales, and Membership Management System. The system is designed to address these issues by integrating inventory management, sales processing, customer membership, reporting, and online payment recording into a centralized platform. The researchers also identified the key system features, including product management, barcode-based product identification, sales transaction processing, QR code-based membership, and inventory tracking. Additionally, the necessary tools, timeframe, and resources for system development were determined during this phase to ensure the successful completion of the project.

System Design Phase

In this phase, the researchers designed the overall structure and interface of the proposed Web-Based Inventory, Sales, and Membership Management System for Boss Rap Motor Shop. This includes the development of user interfaces such as the Login Page, Dashboard, Inventory Module, Sales Module, Membership Module, and Reporting Module. The system design was created to ensure ease of use, allowing users to efficiently navigate and operate the system.

The system interface was structured to provide a clear and organized layout for all users, ensuring that each function of the system is easily accessible and understandable. This supports efficient operation of daily business processes such as inventory tracking, sales recording, and customer management.

The database design was also established to properly store and manage all essential data needed in the system. This ensures that information is organized, consistent, and easily retrievable for business operations.

Database Design

The database was designed to store and manage all essential information required in the operation of Boss Rap Motor Shop. It ensures that all records are centralized, secure, and updated in real time.

The system stores the following types of information:

- **Product Information** – includes product name, description, category, price, and stock quantity
- **Sales Transactions** – includes transaction date, purchased items, quantities, total amount, and payment details
- **Customer Membership Information** – includes customer name, contact number, address, membership ID, and QR code
- **User Accounts** – includes system users such as the shop owner and employees, along with their login credentials and roles

This centralized database allows accurate monitoring of business operations and ensures proper tracking of inventory, sales, and customer records.

System Modules

The system is composed of several functional modules, each designed to handle specific business processes of Boss Rap Motor Shop. These modules work together to ensure efficient management of inventory, sales transactions, customer membership, and reporting.

- **Inventory Management Module**

The Inventory Management Module is responsible for managing and monitoring all product-related information in the system. It allows users to add new products, update existing product details such as name, category, and price, and monitor stock levels. All product records are stored in a centralized database, where each item is assigned a unique identifier for easy tracking. In operation, the user inputs product information into the system, including product name, category, price, and quantity. Once saved, the system stores the data and makes it available for use in other modules, particularly during sales transactions. When a product is sold or manually updated, the system automatically adjusts the stock level in real time. This ensures that inventory records remain accurate and up to date. Through this module, the system minimizes errors associated with manual tracking and improves the efficiency of inventory monitoring.

- **Sales Management Module**

The Sales Management Module handles the recording and processing of all sales transactions within the system. It allows users to select products, compute total amounts, record transactions, and automatically update inventory. The module supports both walk-in and online transactions, ensuring that all sales are recorded in a centralized system. Additionally, the Sales Management Module supports multiple payment methods to accommodate different customer preferences, including cash payments and digital payments such as e-wallets or online banking. This allows the system to handle both traditional and modern transaction types within a single platform. During a transaction, the user selects products either by scanning their barcodes or by manually searching the product list. Once selected, the system retrieves product details such as name, price, and available stock, and adds them to the transaction list. The system then automatically calculates the total amount. After confirming the transaction, the system records the sale and deducts the corresponding quantity from the inventory. This process ensures accurate and efficient transaction handling while reducing human error and improving overall speed.

- **Barcode-Based Product Identification**

To improve efficiency and accuracy in product selection, the system integrates barcode scanning technology. Each product is assigned a unique barcode that is linked to its record in the database. When a barcode is scanned during a transaction, the system automatically retrieves the corresponding product information and adds it to the transaction list. This eliminates the need for manual data entry, reducing processing time and minimizing errors. The system also includes a manual search option as a fallback in case the barcode cannot be scanned. The use of barcode technology enhances transaction speed, ensures accurate product identification, and supports real-time inventory updates.

- **Membership Management Module**

The Membership Management Module is designed to register and manage customer information while supporting customer retention strategies. It enables the system to store customer details, track purchase history, and identify repeat or high-value customers. To become a member, customers are required to provide basic information such as full name, contact number, address, and valid identification. Once registered, the system creates a unique membership record for each customer and stores it in a centralized database. This allows easy retrieval and management of customer information during transactions. During system use, customer membership can be accessed to link transactions to specific individuals, allowing the system to maintain a record of purchase

history. This supports better customer service and enables the business to analyze buying behavior for marketing and retention purposes.

- **QR Code-Based Membership Identification**

The system incorporates QR code technology to improve customer identification and transaction efficiency. Upon registration, each member is assigned a unique QR code that can be printed or stored digitally.

During transactions, the QR code is scanned to instantly retrieve customer information from the system. This process allows purchases to be automatically linked to the customer's account and updates their transaction history in real time. As a result, the system reduces manual data retrieval, minimizes processing time, and improves accuracy in identifying customers. This feature also supports the tracking of customer purchasing behavior, enabling the shop to identify repeat and high-value customers.

- **Reporting Module**

The Reporting Module is responsible for generating reports that support business monitoring and decision-making. It provides users with access to important data such as sales performance, inventory status, and customer activity. The system collects data from various modules and processes it to generate organized reports. Users can request reports based on specific criteria, such as date range or type of transaction. Once generated, the reports can be viewed on the system or printed for documentation purposes. This module eliminates the need for manual report preparation and ensures that business data is accurate, consistent, and readily available. Through this module, the shop owner can effectively monitor daily operations, evaluate performance, and make informed decisions based on real-time data.

Development Phase

In this phase, the researchers began the actual development of the system using appropriate programming languages and database technologies. The designed features from the previous phase were translated into functional modules, including inventory management, sales management, membership management, reporting, and online payment recording.

The researchers developed the user interface components such as the login page, dashboard, and module interfaces for inventory, sales, and membership. The system was integrated with a database to store and manage all records in real time. Key functionalities such as barcode scanning for product identification and QR code generation for membership cards were also implemented to improve efficiency and accuracy.

Additionally, the system was designed to automatically update inventory levels after each transaction and generate reports for sales performance, inventory status, and customer activity. This phase ensures that the system operates according to the planned design and meets the requirements of the study.

Month	January				February				March				April				May				June				July				August				September				October				November			
Activity																																													
Planning																																												
a. Brainstorming																																												
b. Title Planning																																												
c. Requirements Gathering																																												
d. Client Searching																																												
System																																												
a. Functional Mapping																																												
b. System UI Desining																																												
c. Database Design																																												
Development																																												
a. Coding																																												
Testing																																												
a. Unit Testing																																												
b. Integration Testing																																												
c. System Testing																																												
d. User Acceptance Testing																																												
Deployment																																												
a. System Deployment																																												
Review																																												
a. Performance Review																																												
b. Code Review																																												

Resources

HARDWARE LAPTOP COMPUTER

Hardware Components	Specifications
Processor (CPU)	AMD Ryzen 5 5500U with Radeon Graphics (2.10 GHz)
RAM	16.0 GB
Graphic Processing Unit	Integrated graphics processing unit (GPU), AMD Ryzen 5 5500U processor.
Storage	512 GB
System Type	64-bit operating system, x64-based processor

SMART PHONE

Hardware Components	Specifications
IOS SMART PHONE	Apple A13 Bionic
RAM	4 GB
STORAGE	128 GB

Testing Phase

In this phase, the completed web-based system underwent rigorous functional, integration, usability, and security testing to verify operational dependability. Functional testing evaluated the core business logic, including real-time stock deduction upon POS transaction completion, barcode scanner input recognition, member QR code scanning and profile retrieval, points accumulation, and promo discount calculations. Integration testing verified the seamless connection between the React frontend, the Supabase PostgreSQL database, and the Semaphore SMS Gateway API. Usability evaluations were conducted with shop staff to ensure that the POS counter terminal and customer dashboard feature intuitive layouts and responsive touch/keyboard navigation. Security validation confirmed that role-based route guards prevent unauthorized access to administrative functions and that customer contact data is handled securely.

Deployment Phase

In this phase, the system was deployed to a production cloud hosting environment via Netlify, directly integrated with the live Supabase cloud database instance. System user accounts were provisioned with appropriate role permissions for the shop owner (Admin), counter cashiers (Employee), and delivery logistics personnel (Rider). Hardware peripheral integration was completed at the physical shop counter in Baliuag, Bulacan, including configuration of USB/wireless barcode scanners and receipt printers. A comprehensive orientation and user training session was conducted for the shop owner and personnel to familiarize them with product stock-in procedures, POS cashiering, barcode scanning workflows, QR membership issuance, and rider delivery status tracking.

Review Phase

In the final phase, the system's operational performance was evaluated during active store operations. Transaction throughput, stock count accuracy, and data consistency were monitored to ensure that all daily sales and stock intakes synchronized without latency or data loss. User feedback was collected from the shop owner, counter staff, and regular customers to evaluate practical improvements over previous manual methods. Recommendations for continuous enhancements—such as additional payment gateways and expanded inventory analytics—were documented to support

future system upgrades and regular database maintenance.

Software Components

Software Category	Component / Technology Specification
Integrated Development Environment (IDE)	Visual Studio Code 1.90+ with TypeScript and ESLint toolchains
Frontend Framework & Language	React 19, TypeScript 6.0, Vite 8.0 (Modern Single-Page Application)
User Interface & Visualization	Vanilla CSS Design System, Lucide React Iconography, Recharts Library
Cloud Backend & Database	Supabase Platform (PostgreSQL 15 Relational DB with Foreign Keys & Triggers)
Cloud File Storage	Supabase Cloud Storage (Bucket: 'Files' for E-receipts & Product Media)
SMS Notification Gateway	Semaphore SMS RESTful API Gateway (Sender: TranslertPH)
Hardware Scanner & Symbology	USB/Wireless Keyboard Emulation Barcode Scanner (EAN-13, Code 128)
QR Code Generation Engine	Dynamic QR Code Generator API for Digital Membership Identification
Operating System Compatibility	Windows 10 / Windows 11 (Store Counter POS & Administrative Desktop)
Web Browser Compatibility	Google Chrome 120+, Microsoft Edge, Mozilla Firefox, Apple Safari

Requirements Analysis

Who – The People Involved

The primary stakeholders and users of the system include the shop owner/administrator, store counter employees (cashiers), customers (registered members and walk-in patrons), and delivery riders. The shop administrator exercises comprehensive authority over product catalog pricing, stock inventory levels, supplier purchase orders, promo campaigns, and financial performance reports. Store employees (cashiers) utilize the dedicated POS station to process sales, scan product barcodes, verify member QR cards,

and issue receipts. Customers interact with the system to browse available motorcycle spare parts and servicing menus, register for digital membership, present their digital QR code card for tiered loyalty points, and submit warranty replacement requests. Delivery riders access the dedicated rider portal to review assigned deliveries, update parcel progress, and trigger automated SMS alerts to recipients.

What – The Business Activity

The business activity centers on digital sales processing, real-time parts inventory monitoring, supplier restocking, and customer relationship management. The system records in-store walk-in purchases as well as orders originating from online/Facebook inquiries. Each completed transaction automatically decrements product quantities from inventory, awards loyalty points based on store-configured rules, and archives transaction records with payment verification codes. In addition, the platform manages supplier purchase orders to streamline stock replenishment, provides automated return/replacement claim workflows, logs customer inquiries, and compiles real-time sales summaries.

Where – The Environment Where the Work Takes Place

The physical environment is Boss Rap Motor Shop located in Baliuag, Bulacan, where in-store parts retailing, motorcycle repair servicing, and cashiering take place. The digital system operates in a secure cloud web environment accessible via standard desktop and mobile browsers over HTTPS. The shop counter utilizes desktop or laptop computers equipped with barcode scanners and receipt printers for counter POS operations, while the shop owner, delivery riders, and customers can access the web application remotely from smartphones and personal devices.

When – The Timing of the Process

Walk-in sales transactions and counter servicing occur during standard motor shop operating hours (daily from 8:00 AM to 6:00 PM). The web platform, product catalog, and online customer inquiry modules operate continuously (24/7), allowing customers to browse spare parts, submit purchase orders, and verify member loyalty status at any time. Delivery status updates and SMS alerts are triggered synchronously during transit,

while administrative inventory restocking and financial report generation are performed on demand or during weekly audit cycles.

How – Current Procedures and System Solution

Under the traditional procedure, transactions and stock counts were recorded manually in notebooks and loose sheets, leading to data loss, calculation inaccuracies, and unorganized customer records. Orders placed through Facebook Messenger required staff to manually check physical stock, calculate totals on desktop calculators, and manually write paper receipts. The proposed web application replaces these fragmented practices with an integrated digital workflow. Product barcodes are scanned directly into the POS terminal; totals, taxes, and member discounts are automatically computed; inventory balances update instantaneously across all connected screens; official receipts are printed digitally; and automated SMS notifications keep delivery customers informed in real time.

Requirements Documentation

Stakeholder Interview Transcript

To formulate accurate operational specifications and identify the practical challenges of the business, the proponents conducted a comprehensive semi-structured interview with the management of Boss Rap Motor Shop. The transcript of the initial operational assessment interview is documented below:

Interviewee: Proprietor/StoreManager(BossRapMotorShop)
Interviewers: Vladimmhir B. Canchino, Aljae Dj. Cruz, Mohammad Hamdie T. Langco, Don Janken B. Teodoro

Meeting Place: Boss Rap Motor Shop, Baliuag, Bulacan
Date Conducted: October 14, 2026

Proponents: Magandang araw po! Kami po ay mga mag-aaral ng Bachelor of Science in Information Technology mula sa STI College Baliuag. Nais po naming magtanong ukol sa kasalukuyang sistema ng inyong operasyon sa Boss Rap Motor Shop.

Shop Owner: Magandang araw din sa inyo. Sige, ayos lang, handa akong sumagot sa mga

katanungan ninyo ukol sa aming tindahan.

Proponents: Sir, paano po ninyo kasalukuyang naitatala ang inyong mga imbentaryo ng motorcycle parts at accessories? Gumagamit po ba kayo ng computerized software?

Shop Owner: Sa ngayon, mano-mano pa ang aming talaan. Gumagamit kami ng notebook o logbook para isulat ang mga dumarating na piyesa mula sa supplier at ang mga nababawas kapag may bumibili. Minsan, kapag abala sa talyer o maraming customer sa counter, hindi agad naisusulat kaya nagkakaroon ng discrepancy o kulang sa bilang kapag nag-imbentaryo.

Proponents: Paano naman po kapag may nagtatanong na customer sa presyo o kung available ang isang part tulad ng brake pads, spark plug, o gulong?

Shop Owner: Kailangan pang tingnan sa istante o kaya hanapin sa lumang listahan kung magkano ang benta. Kapag bago ang tauhan o mekaniko, natatagalan bago masagot ang customer dahil hindi kabisado ang bawat presyo.

Proponents: Mayroon din po ba kayong mga transaksyon online o sa pamamagitan ng social media?

Shop Owner: Oo, marami kaming natatanggap na inquiries at orders sa aming Facebook page. Nagtatanong sila kung may stock ng pyesa, tapos nagpapadala kami ng litrato. Kapag bibili sila, pinipili nila kung ipapa-pickup sa shop o ipapahatid gamit ang third-party delivery o sarili naming rider. Ang bayad minsan ay GCash, minsan cash pagdating.

Proponents: Paano po ninyo namomonitor ang mga orders na iyon at ang mga payments sa GCash?

Shop Owner: Sa cellphone ko lang tinitingnan ang GCash confirmation SMS, tapos manual kong sasabihin sa tauhan na ibalot na ang item. Dahil walang iisang database, minsan nagkakaroon ng kalituhan kung aling bayad ang tumutugma sa aling order, lalo na kapag sabay-sabay ang orders sa maghapon.

Proponents: Mayroon po ba kayong sistema para kilalanin ang mga suki o regular customers, tulad ng membership o loyalty points?

Shop Owner: Wala pa kaming pormal na membership card o points system. Pag suki, binabawasan na lang namin ng konti sa presyo base sa pakikipag-usap. Pero maganda kung may digital membership para kapag nag-scan sila ng QR code o card, may automatic points sila at makikita ang kanilang purchase history.

Proponents: Kung magkakaroon po ng web-based system na may barcode scanner sa counter, automated stock deduction, QR membership cards, at automated SMS alert kapag out for delivery ang order ng customer, makakatulong po ba ito nang malaki sa inyong shop?

Shop Owner: Napakalaking tulong niyan! Mabilis na ang pagbenta sa counter gamit ang barcode scanner, hindi na mawawala ang records dahil nakatago sa cloud database, at

masisiyahan ang mga suki dahil may points sila at may text message kapag paparating na ang kanilang parcel.

Proponents: Maraming salamat po sa inyong oras at sa mga mahahalagang impormasyon na inyong ibinahagi. Gagamitin po namin ito bilang pundasyon sa pagbuo ng sistema.

System Architecture

The proposed Web-Based Inventory, Sales, and Membership Card System for Boss Rap Motor Shop utilize a three-tier web architecture. This structure organizes system components into separate layers, enabling efficient processing, scalability, and maintainability. The system is composed of three main layers: the presentation layer, the application layer, and the database layer.

The **presentation layer** serves as the user interface of the system. It consists of web-based pages accessed through a web browser. The shop owner and employees can log in to the system using internet-connected devices such as computers or laptops. This layer allows users to manage products, record sales transactions, register customer membership information, and access system reports and records.

The **application layer** handles the processing of user requests and contains the core business logic of the system. It processes all inputs submitted through the presentation layer, such as product entries and sales transactions. It is also responsible for executing system functions including inventory updates, sales computation, and data validation before storing or retrieving information from the database.

The **database layer** is responsible for the secure storage and management of all system data. This includes product inventories, sales records, membership information, and user accounts. A database management system is used to ensure data integrity, security, and efficient retrieval and updating of records when required.

Overall, the three-tier architecture allows the system to separate user interface, processing logic, and data storage. This improves system performance, enhances security, and ensures better organization of data. It also enables efficient management

of inventory, sales, and membership records within the system.

Simple Use Case Diagram Structure

The Simple Use Case Diagram illustrates the interaction between the users and the Web-Based Inventory, Sales, and Membership Card System for Boss Rap Motor Shop. It identifies the main actors of the system, which include the shop owner and employees, and shows the key functions they can perform within the system.

The main use cases of the system include managing inventory, recording sales transactions, and handling customer membership registration. These use cases demonstrate how users interact with the system to perform essential business operations. The diagram also shows that all processes are accessed through a centralized web-based platform, ensuring organized and efficient workflow.

System Interface Design

The System Interface Design presents the layout and visual appearance of the Web-Based Inventory, Sales, and Membership Card System for Boss Rap Motor Shop. It illustrates how users interact with the system and how the system functions are organized across different pages.

The system is designed to be simple, user-friendly, and easy to navigate. Its interface enables the shop owner and employees to efficiently access key functions such as inventory management, sales transactions, and membership registration. Each page is structured with a clear and consistent layout to ensure that users can easily locate the information and features they need.

The system consists of several important pages, including the Login Page, Dashboard, Inventory Page, Sales Page, and Membership Page. The Login Page provides secure access to the system. The Dashboard displays an overview of system activities and summary data. The Inventory Page is used for managing product stocks, while the Sales Page handles transaction recording and monitoring. The Membership Page manages the registration and records of customer memberships.

Overall, the interface design ensures that users can perform their tasks efficiently while maintaining clarity, consistency, and ease of use throughout the system.

User Levels of the System

The proposed Web-Based Inventory, Sales, and Membership Management System for Boss Rap Motor Shop includes two primary user levels: the **Admin** and the **Customer**. Each user level is assigned specific functions and access permissions to ensure proper system operation, data security, and efficient management of business processes.

Admin User Level

The admin serves as the primary authorized user responsible for managing and monitoring the overall operations of the system. This user level has full access to the major system modules and administrative functions.

The admin can perform the following functions:

- Manage product inventory by adding, updating, deleting, and categorizing products
- Monitor stock levels and inventory movement in real time
- Use barcode scanning functionality for efficient product identification during inventory handling and sales transactions
- Record and process walk-in and online sales transactions
- Manage customer membership records and generate QR code-based membership cards
- View and monitor customer transaction history and membership information
- Manage user accounts and control system access permissions
- Record and monitor online payment transactions such as e-wallet and online banking payments
- Generate and print reports related to sales performance, inventory status, customer activity, and product movement
- Monitor overall system activities through the dashboard interface
- Manage the membership loyalty system by configuring point conversion rules and customer reward settings
- Create, update, and delete promotional offers and discount campaigns within the system
- Control the implementation of promos, including eligibility conditions, discount

values, and promotional durations

The admin user level ensures that business operations are properly managed, monitored, and maintained within the system.

Customer User Level

The Customer user level is intended for registered members of Boss Rap Motor Shop who interact with the system primarily for membership and customer-related services. The Customer can perform the following functions:

- Register as a member by providing required personal information
- Log in securely using their registered account credentials
- View available motorcycle products and services offered by the shop
- Access and present their QR code-based membership identification during transactions
- View and monitor their transaction history and membership information
- Track earned membership points and loyalty-related records
- Send inquiries or messages through the contact page of the system
- Submit requests for product return or replace, subject to the policies and approval of the shop administrator

The Customer user level is designed to improve customer engagement, support customer retention, and provide a more organized membership experience within the system.

Employee / Cashier User Level

The Employee / Cashier user level is designated for store counter staff and mechanics who manage daily counter transactions at the physical shop. This user level can perform the following functions:

• Access and operate the dedicated Staff Point of Sale (POS) cashiering terminal station
• Execute rapid product item lookup using the integrated hardware barcode scanner emulation
• Scan and verify customer digital QR membership cards to automatically retrieve member profiles and calculate loyalty tier discounts
• Process customer payments across multiple channels: Cash, E-Wallets (GCash / Maya), and Bank Transfers
• Enter and verify electronic payment reference numbers for cashless financial record-keeping
• Generate, preview, and print standardized official sales receipts for walk-in counter customers
• Inspect real-time product stock counts and inventory availability to assist in-store customer inquiries

Delivery Rider User Level

The Delivery Rider user level is configured for shop dispatch and delivery logistics personnel who manage customer parcel fulfillment. Riders access the mobile-

responsive logistics portal (/rider/deliveries) from smartphones or handheld tablets. This user level can perform the following functions:

- Securely log in to the dedicated Rider Logistics Portal with assigned rider credentials
- View assigned delivery orders filtered by status: All, Preparing, Out for Delivery, and Delivered
- Access full delivery fulfillment details, including recipient name, destination address, and interactive Google Maps links
- Update order transit status to 'OUT FOR DELIVERY', which automatically triggers an automated SMS dispatch alert to the recipient
- Update delivery status to 'COMPLETED' upon successful parcel drop-off and payment collection
- Maintain real-time synchronization with administrative sales records to ensure complete delivery accountability

System Interface Modules and Storyboarding

The following figures depict the actual, fully deployed graphical user interfaces of the Web-Based Sales and Inventory System with Membership Management for Boss Rap Motor Shop. Each module is engineered with modern responsive layouts, real-time database synchronization via Supabase, and tailored controls matching the operational workflows of the motorcycle repair and parts retailing enterprise:

Home Page Module

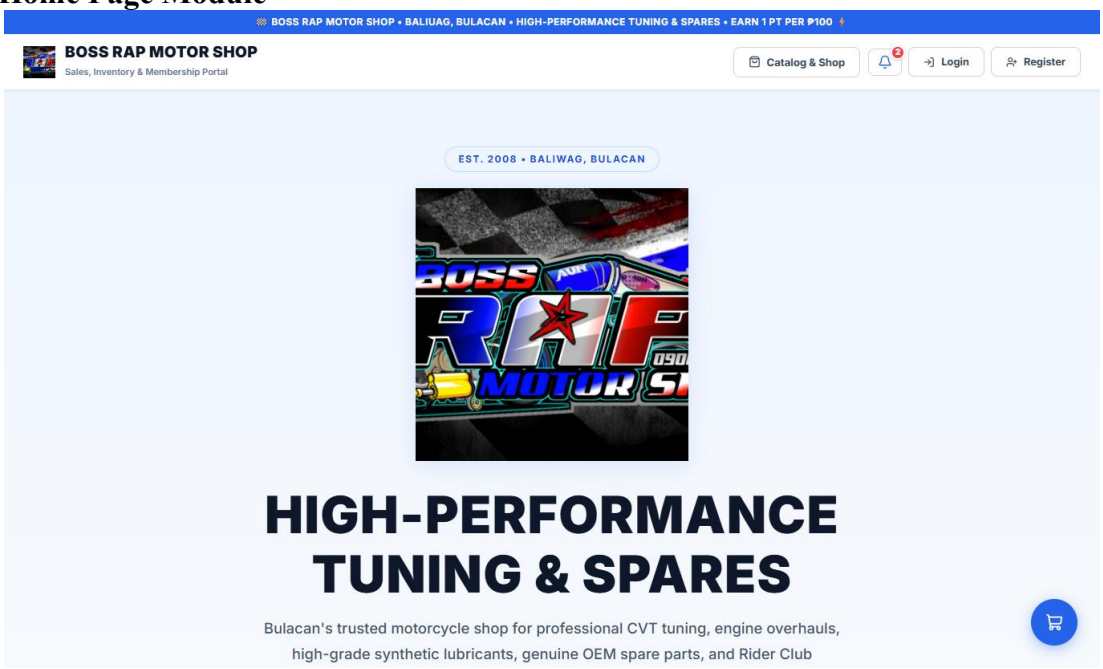


Figure 1: Home Page Module – Displays the main landing interface with the official Boss Rap Motor Shop emblem, operating hours, shop location, quick inventory access buttons, and service offerings.

Online Spare Parts Catalog and Shop Module

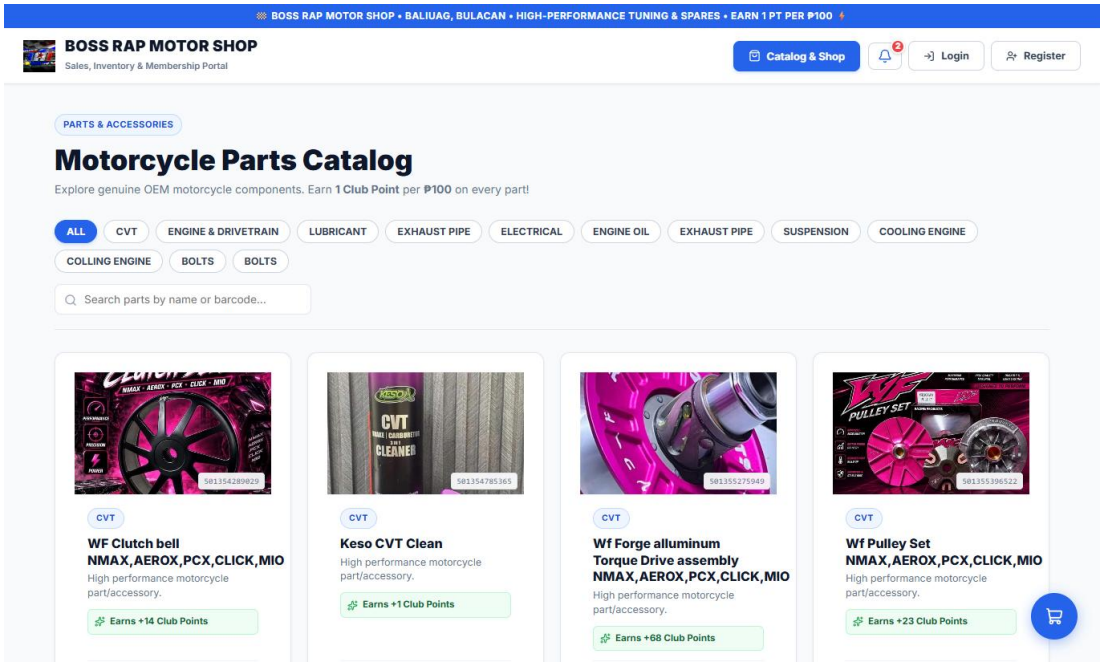


Figure 2: Online Spare Parts Catalog and Shop Module – Enables customers to browse available motorcycle spare parts, filter by category (CVT, Engine, Lubricants, Exhaust), search items by name or barcode, and view point yields.

Maintenance and Servicing Packages Module

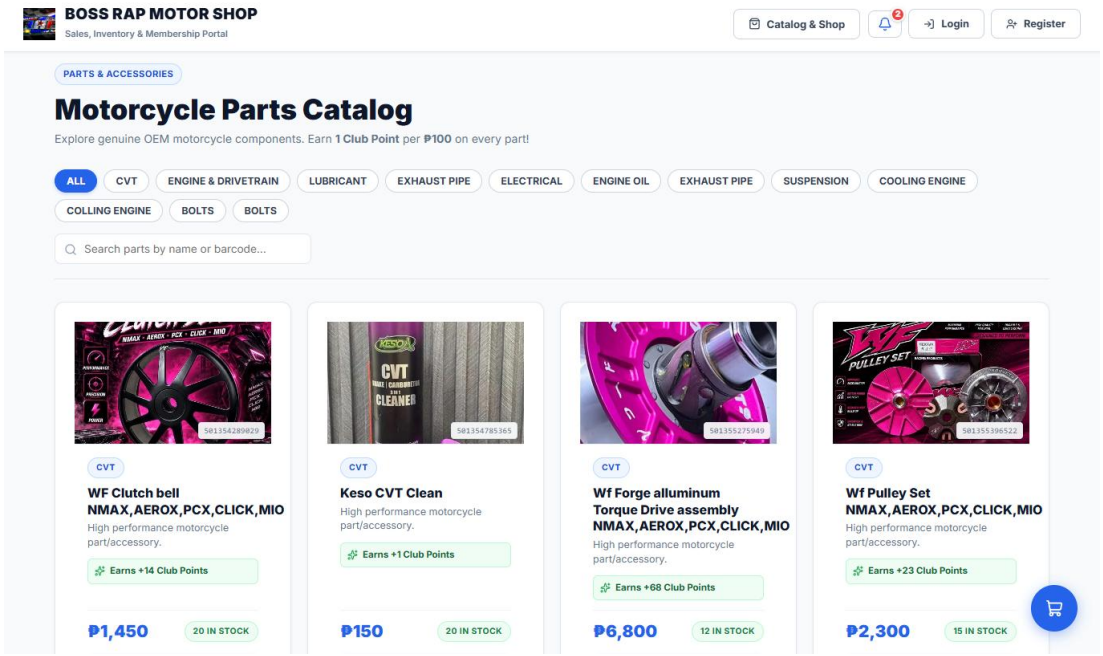


Figure 3: Maintenance and Servicing Packages Module – Presents available motorcycle maintenance services (oil change, tire replacement, CVT cleaning, and general tune-up) with service durations and details.

System Sign-In and Role-Based Authentication Module

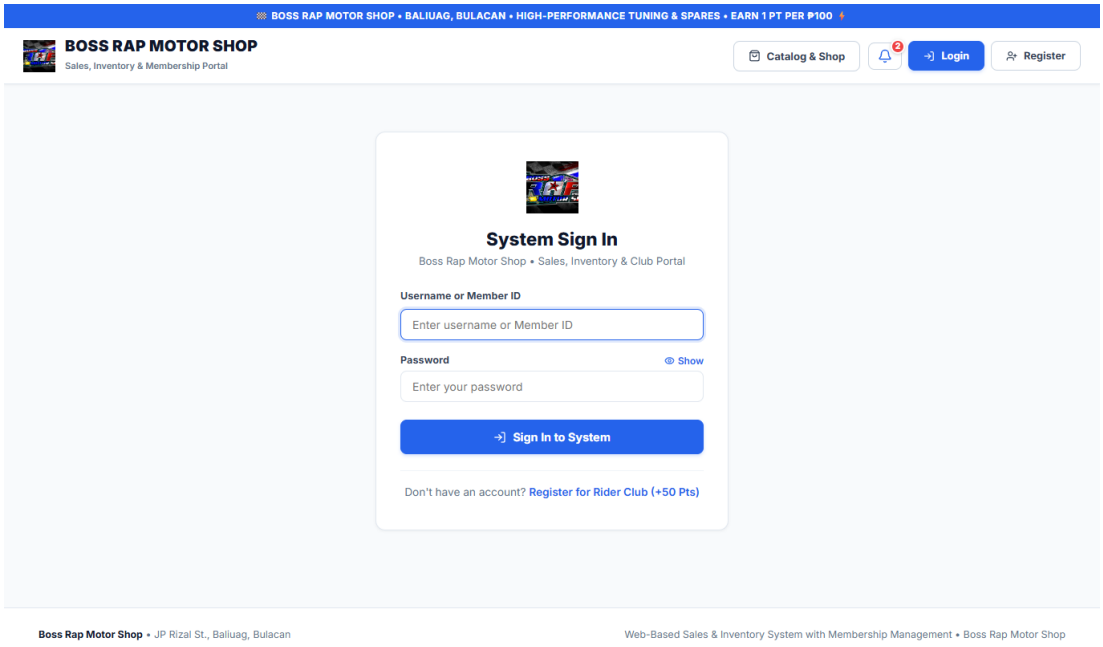


Figure 4: System Sign-In and Role-Based Authentication Module – Provides secure access control and role-based authentication for Admin, Employee Cashier, Delivery Rider, and Customer Member user levels with password visibility toggle.

Customer Member Account Registration Module

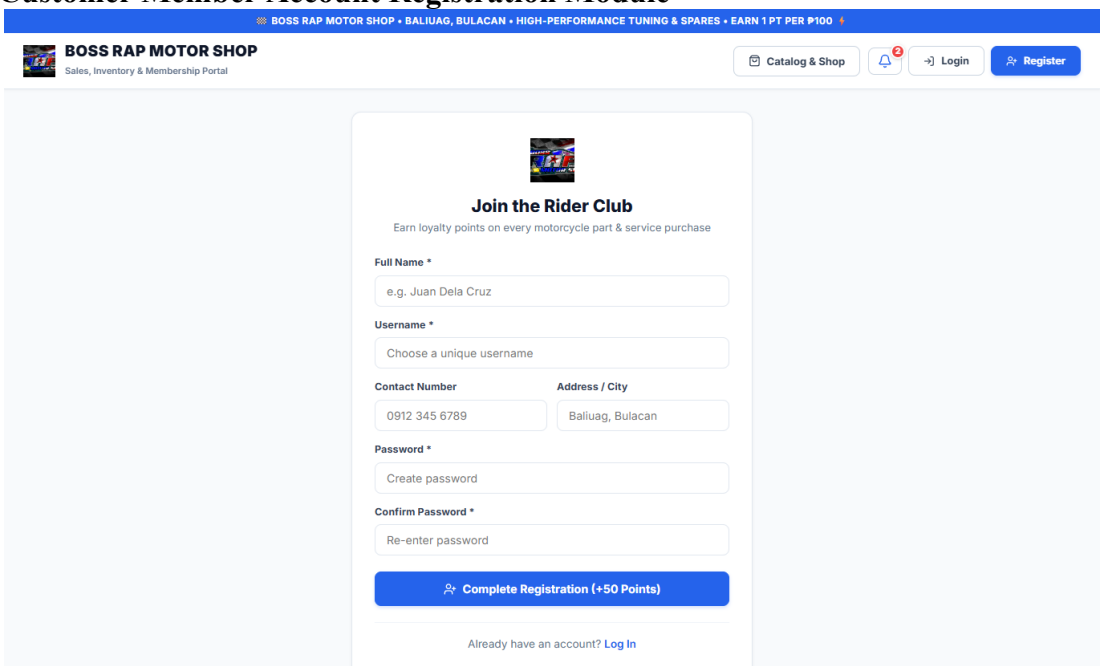


Figure 5: Customer Member Account Registration Module – Allows customers to register for official Boss Rap Rider Club membership, capturing essential contact numbers and delivery address information.

Administrator Dashboard Overview Module

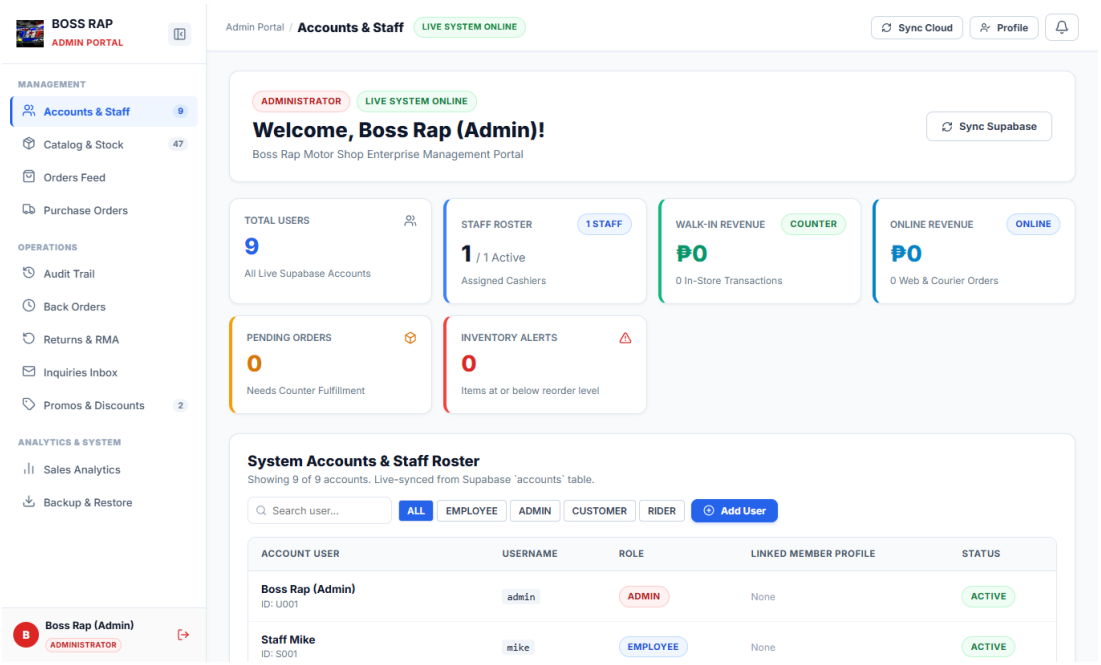


Figure 6: Administrator Dashboard Overview Module – Presents executive business indicators, including total accounts, active cashiers, walk-in revenue, online sales, low stock alerts, and quick inventory shortcuts.

Spare Parts Inventory and Stock Management Module

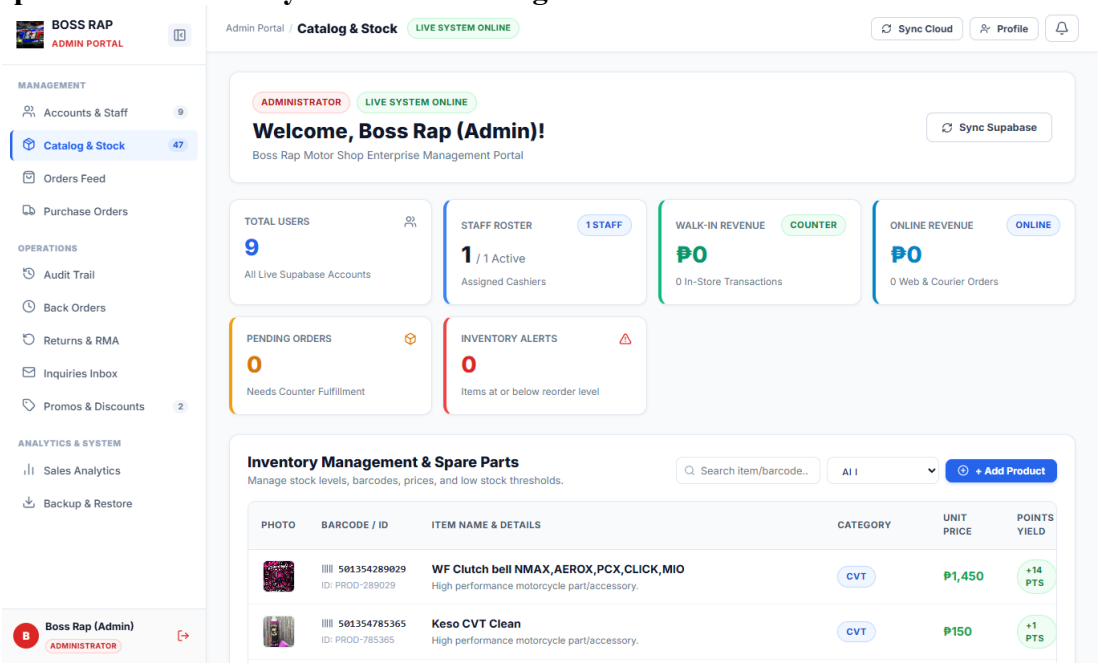


Figure 7: Spare Parts Inventory and Stock Management Module – Enables shop administration to track unit prices, stock levels, item barcodes, low-stock threshold alerts, and perform real-time product updates.

Supplier Purchase Orders (PO) Management Module

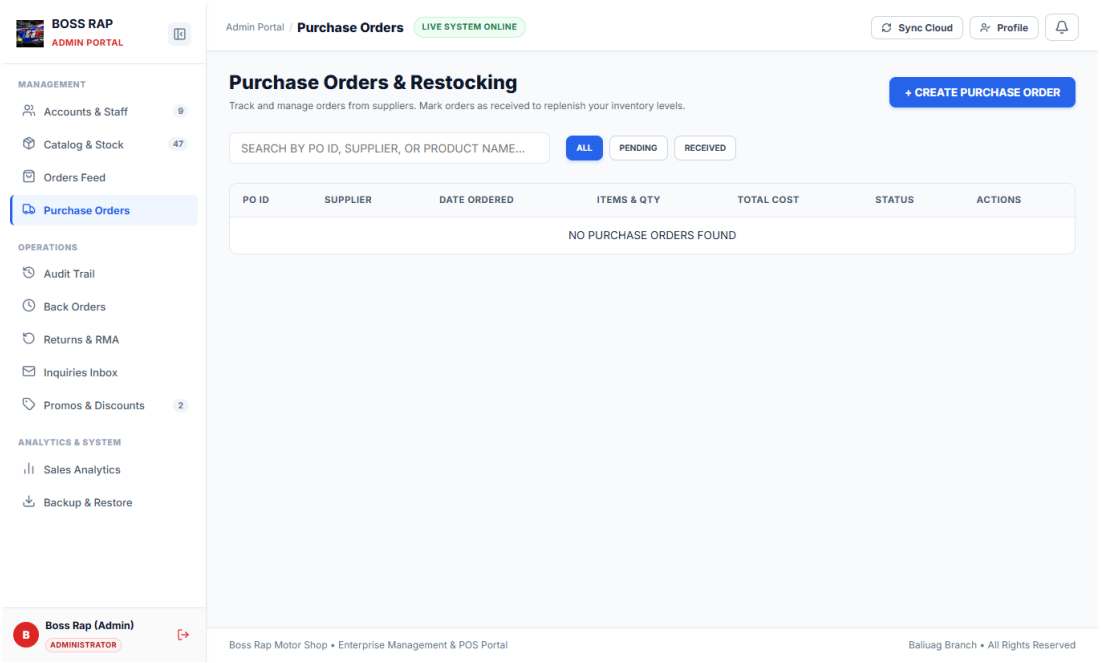


Figure 8: Supplier Purchase Orders (PO) Management Module – Facilitates inventory replenishment by allowing administrators to create purchase orders, monitor supplier deliveries, and verify received stock.

Staff Point of Sale (POS) Cashier Terminal Module

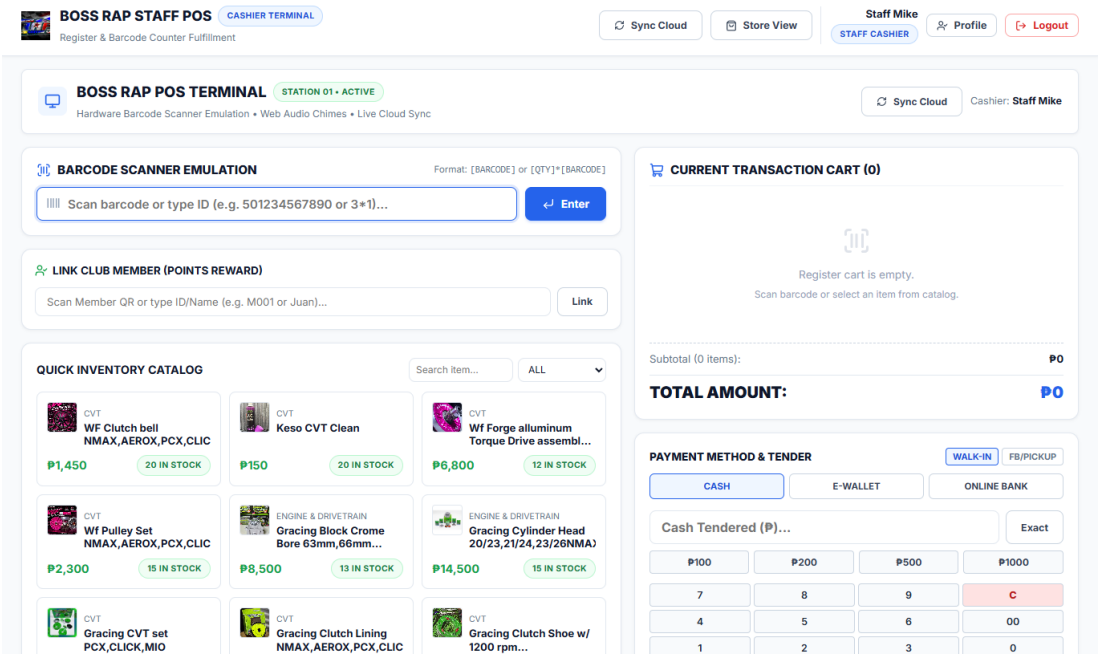


Figure 9: Staff Point of Sale (POS) Cashier Terminal Module – Dedicated counter workstation for staff cashiers featuring hardware barcode scanner emulation, quick-tap parts catalog, customer QR code linking, and cart calculations.

Official Sales Receipt Generation and Printing Module

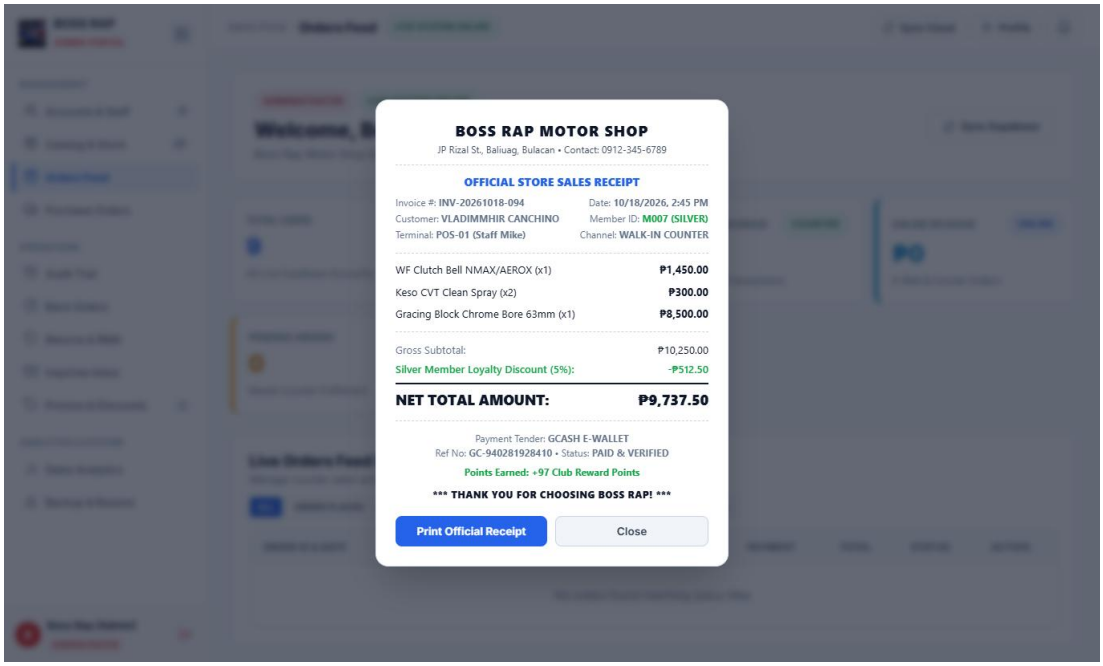


Figure 10: Official Sales Receipt Generation and Printing Module – Automatically generates standardized printable sales receipts displaying transaction timestamp, cashier ID, itemized purchases, member loyalty discounts, and QR code verification.

Orders Feed and Sales Transaction Records Module

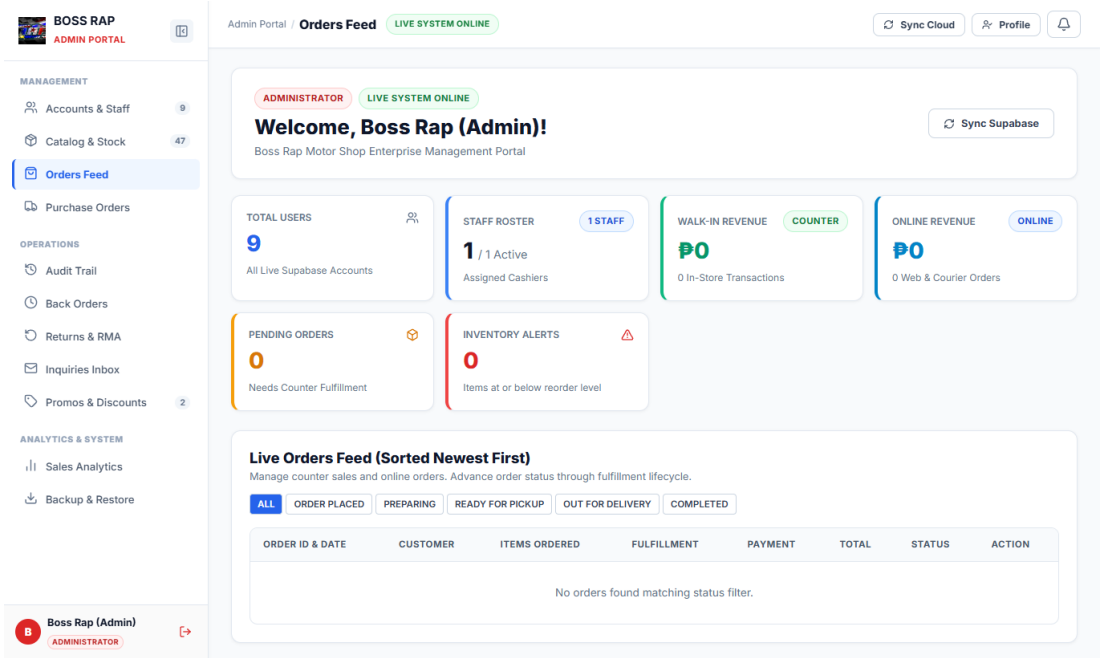


Figure 11: Orders Feed and Sales Transaction Records Module – Centralizes sales transactions across counter POS and online channels, logging invoice details, customer information, and real-time fulfillment statuses.

Online Delivery Fulfillment and Dispatch Tracking Module

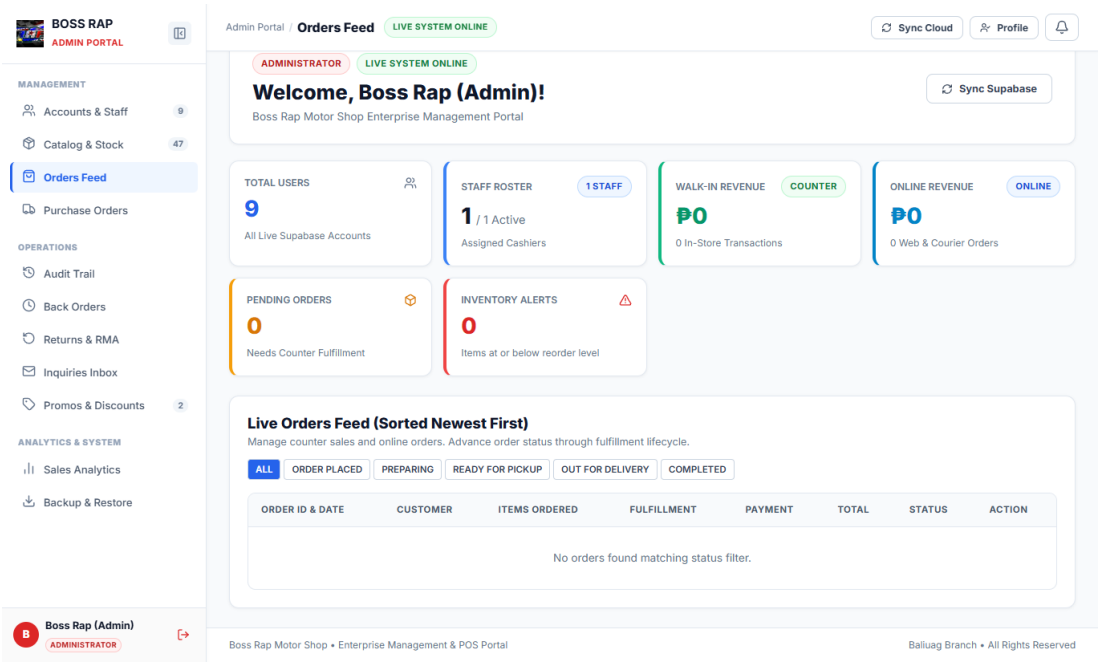


Figure 12: Online Delivery Fulfillment and Dispatch Tracking Module – Displays active delivery orders with destination addresses, courier assignments, status progression (Preparing, In Transit, Delivered), and automated SMS alerts.

Cashless E-Wallet Payment Verification Module

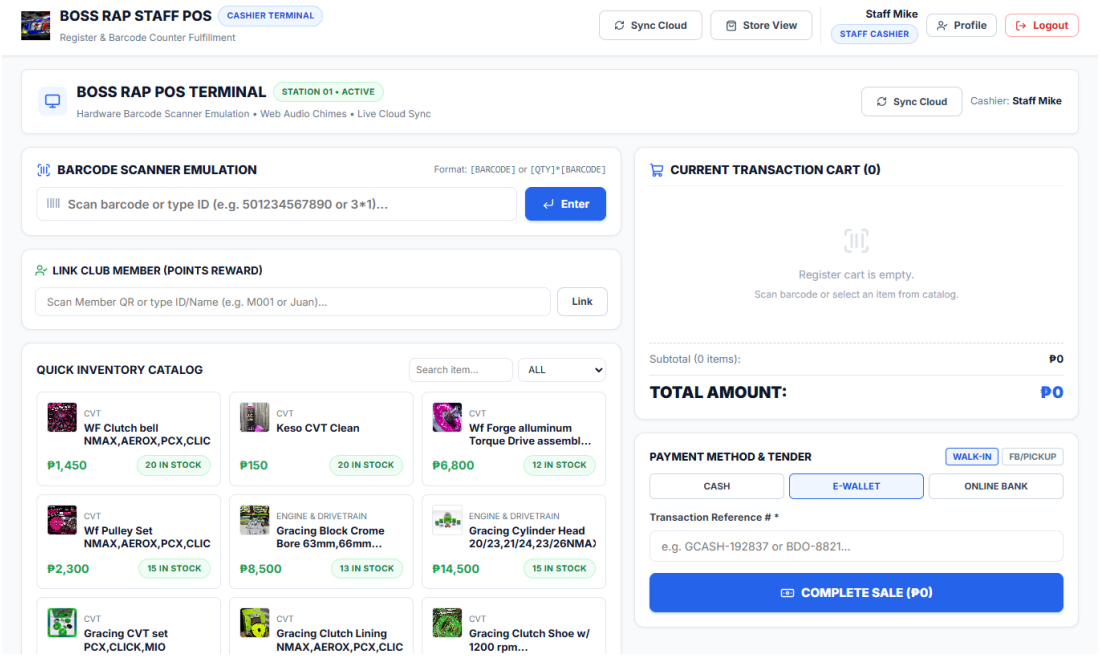


Figure 13: Cashless E-Wallet Payment Verification Module – Supports cashless digital transactions (GCash and Maya), allowing cashiers to record electronic payment reference numbers for financial auditing.

Customer Club Membership Registry and Points Configuration Module

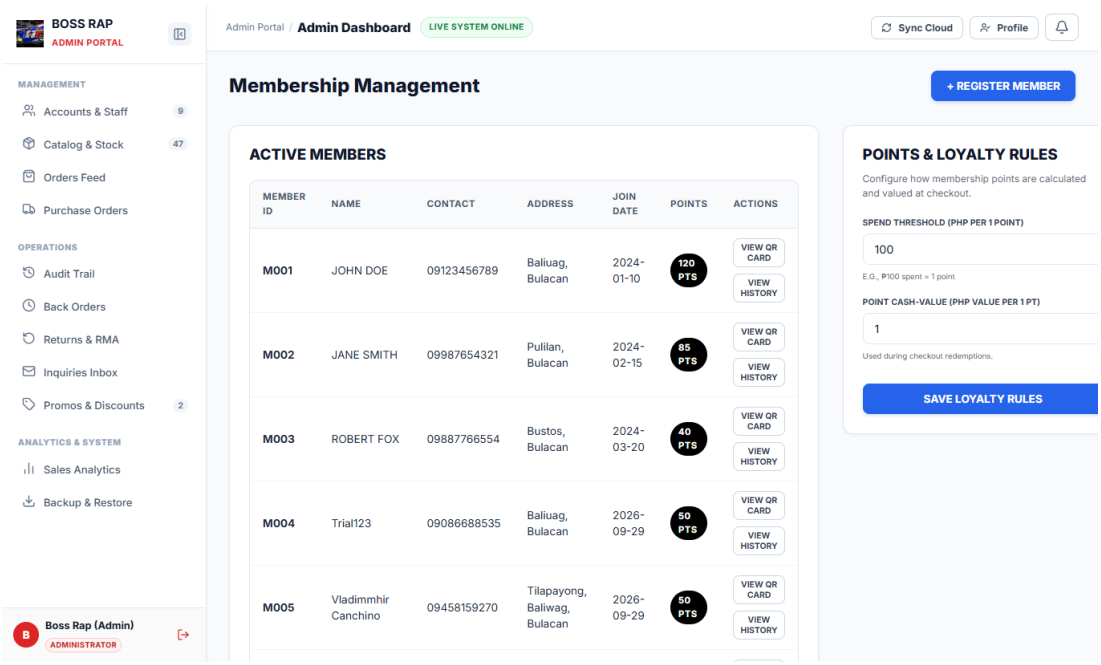


Figure 14: Customer Club Membership Registry and Points Configuration Module – Displays registered customer profiles, accumulated reward points, QR cards, and configurable loyalty rules (spend threshold per point and cash-value conversions).

Promotional Offers and Discount Campaign Management Module

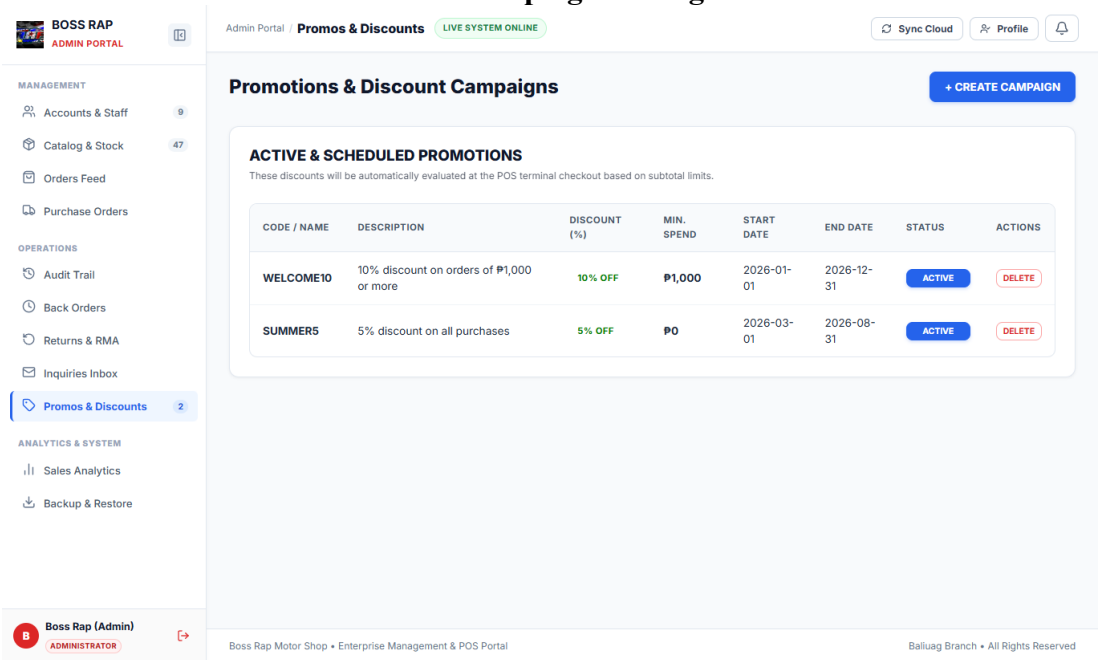


Figure 15: Promotional Offers and Discount Campaign Management Module – Empowers management to create and monitor discount campaigns, promo codes, and special customer incentives applied across POS and online checkouts.

Return, Replacement, and Warranty Claims Management Module

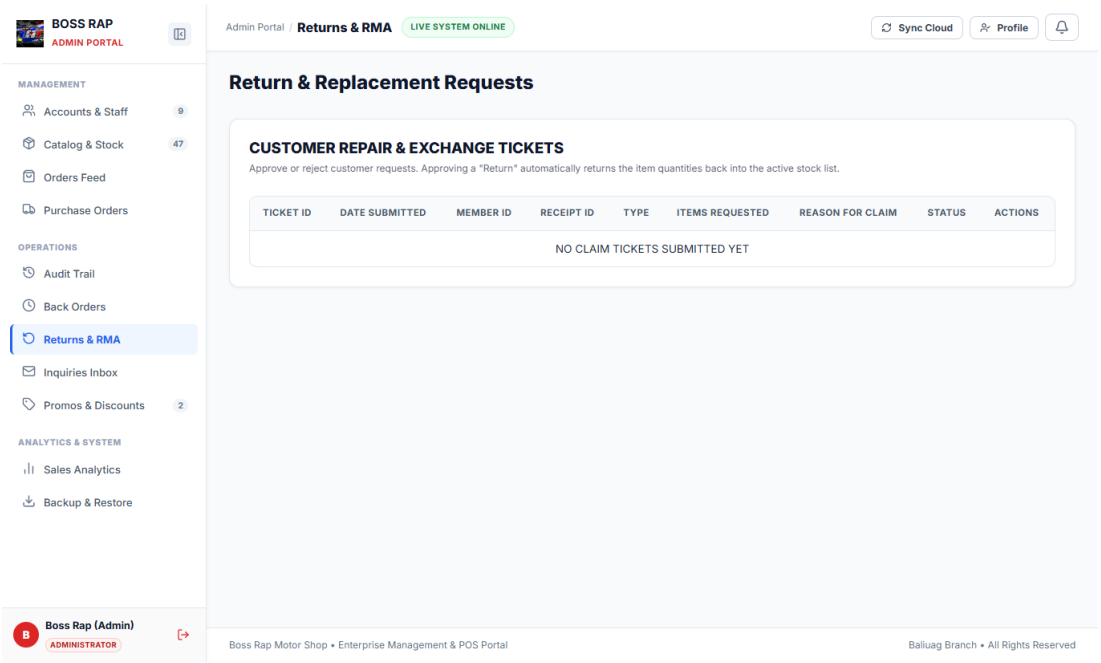


Figure 16: Return, Replacement, and Warranty Claims Management Module – Tracks customer claims for defective parts, supporting upload of proof photos, verification of receipt numbers, and authorization of replacements.

Customer Inquiries and Support Messaging Center Module

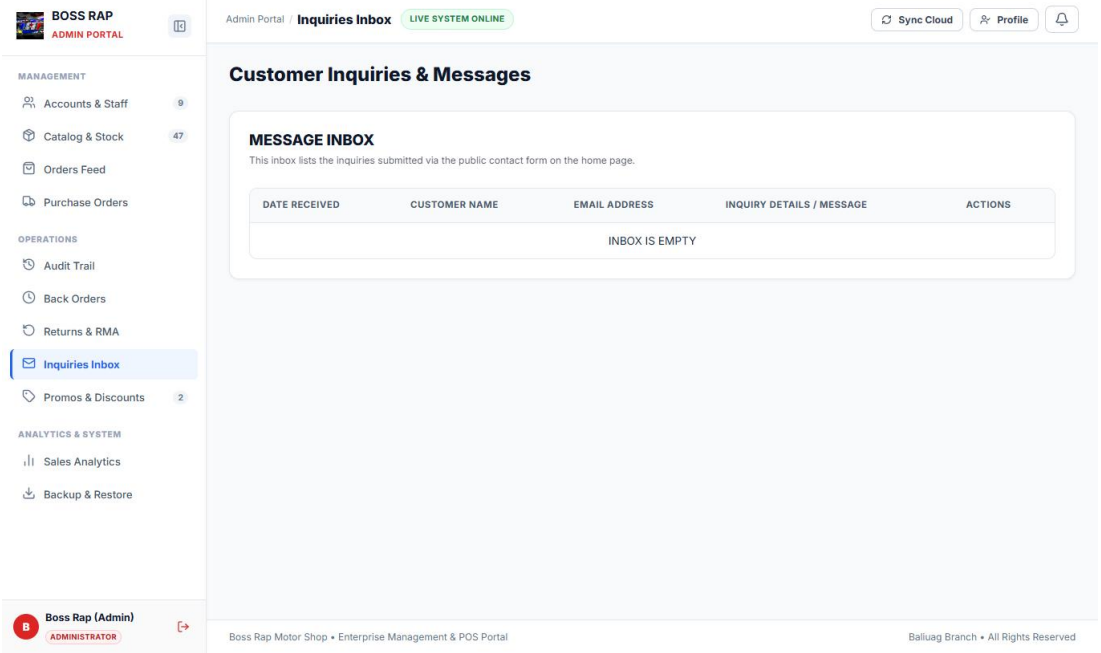


Figure 17: Customer Inquiries and Support Messaging Center Module – Provides a unified inbox for inquiries submitted through the contact portal, enabling management to respond promptly to customer questions.

Customer Member Dashboard and Digital QR Code Card Module

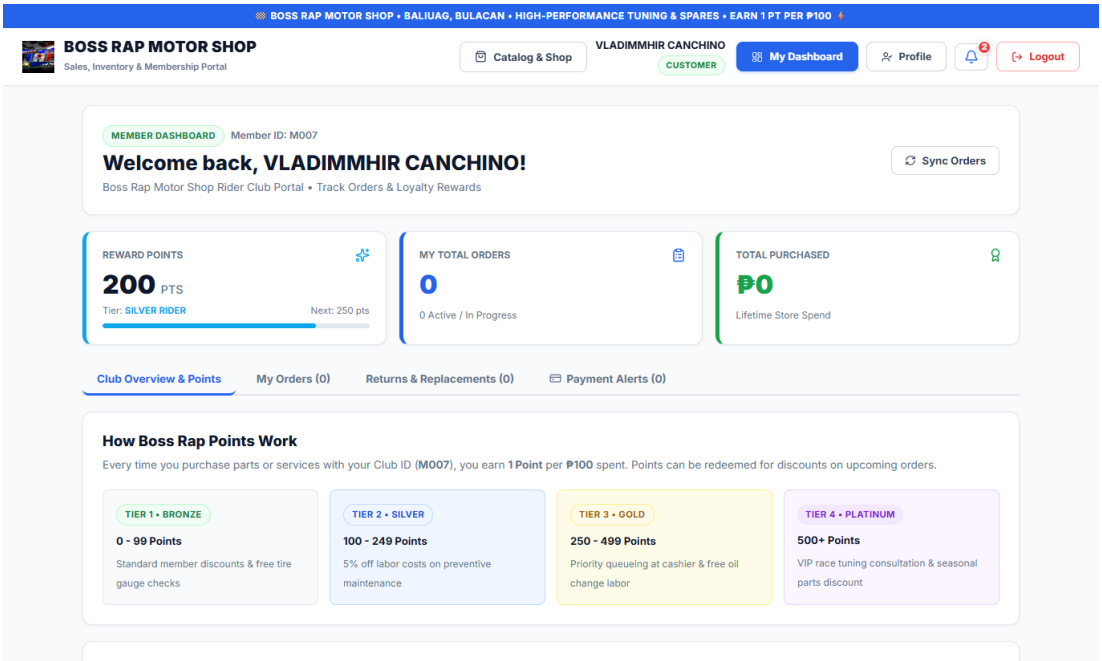


Figure 18: Customer Member Dashboard and Digital QR Code Card Module – Displays customer reward points, current loyalty tier badge (Bronze, Silver, Gold, Platinum), purchase history, and personal digital QR card.

Customer Loyalty Tier Progression and Profile Management Module

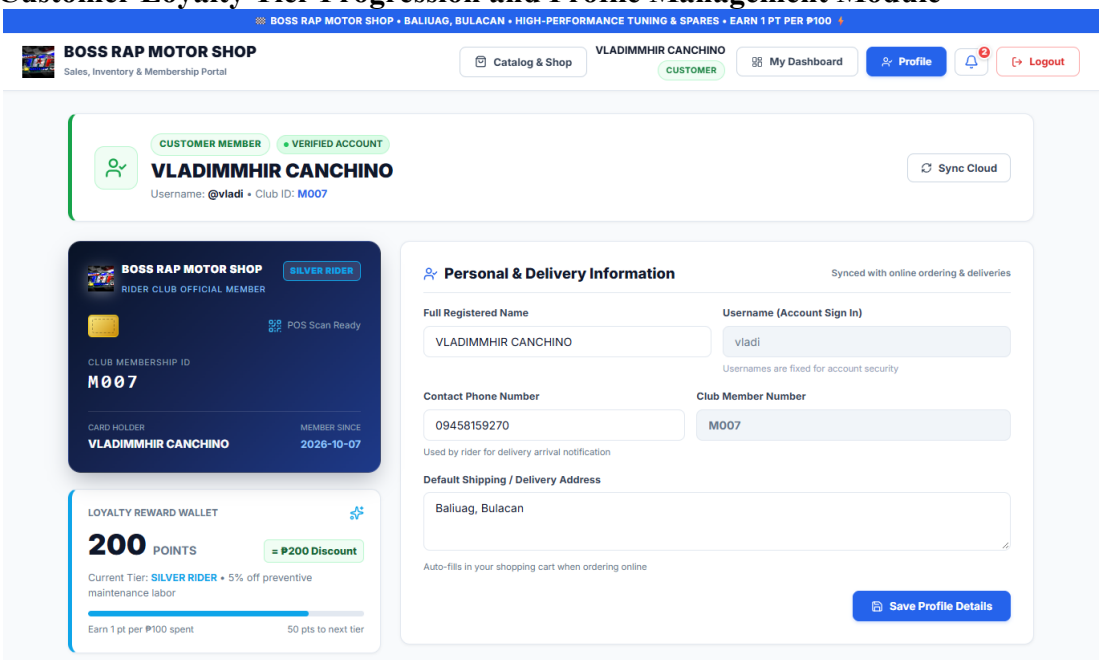


Figure 19: Customer Loyalty Tier Progression and Profile Management Module – Illustrates customer account details, delivery address management, points-to-discount conversion wallet, and milestone tier progression.

Delivery Rider Logistics and Mobile Dispatch Portal Module

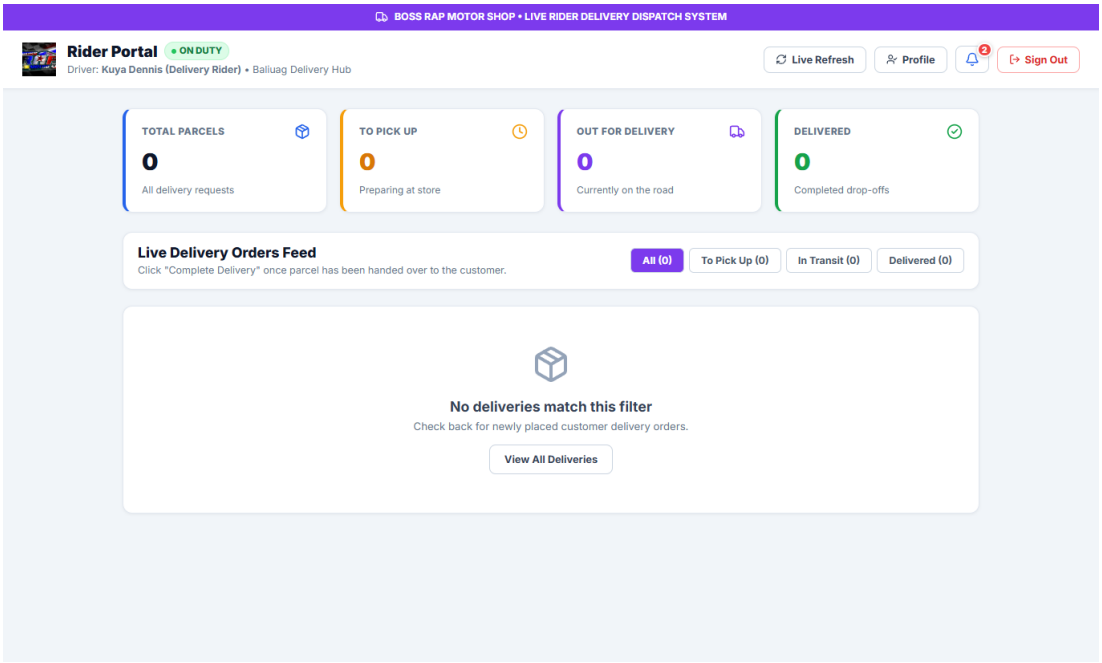


Figure 20: Delivery Rider Logistics and Mobile Dispatch Portal Module – Mobile-responsive delivery portal enabling riders to view assigned parcels, access recipient addresses and Google Maps navigation, and advance transit statuses.

Business Reports, Financial Analytics, and Stock Movement Module

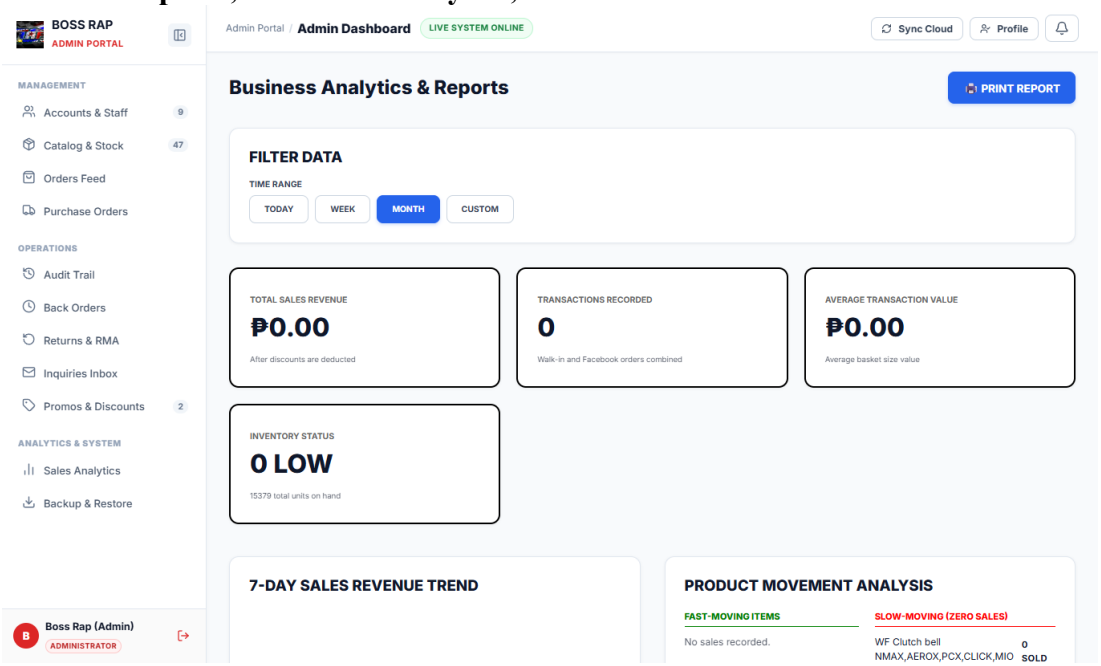


Figure 21: Business Reports, Financial Analytics, and Stock Movement Module – Provides graphical business intelligence, including sales revenue trends, top-selling fast-moving items, deadstock indicators, and printable audit reports.

System User Account Administration and Role Access Governance Module

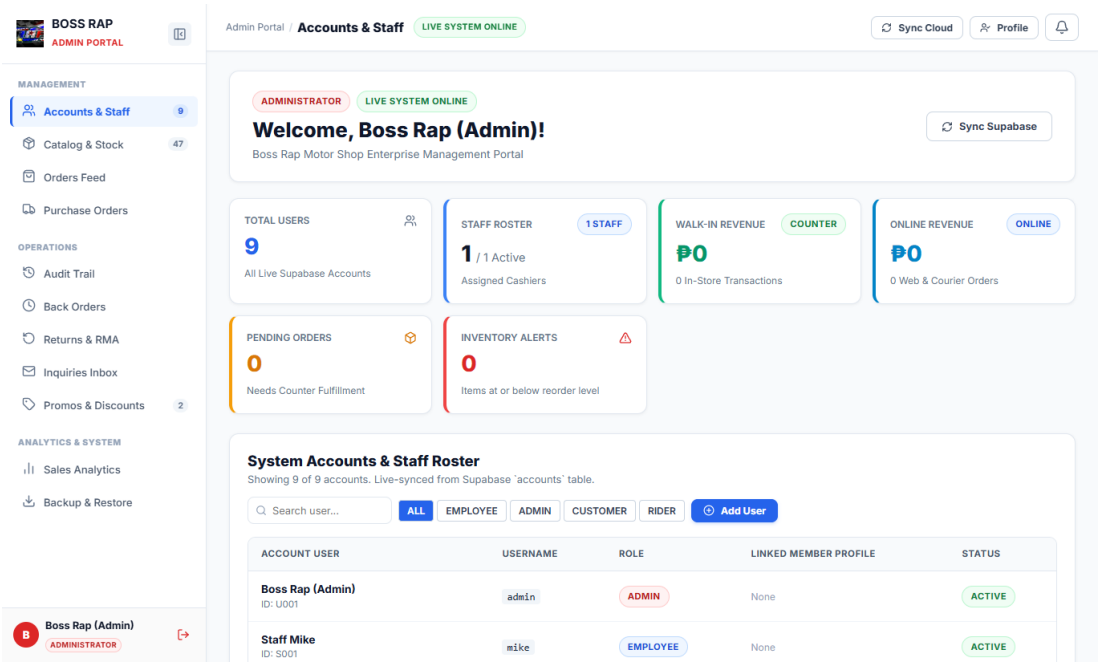


Figure 22: System User Account Administration and Role Access Governance Module – Enables administrators to create staff accounts, update user credentials, manage role assignments, and govern account permissions across the platform.

Design of Software, System, Product, and/or Processes

This section presents the architectural, structural, and procedural design models that govern the development and operation of the Web-Based Sales and Inventory System with Membership Management for Boss Rap Motor Shop. The design encompasses the development life cycle methodology, three-tier web architecture, relational database schema, functional use case interactions, transactional flowcharts, and the Hierarchical Input-Process-Output (HIPO) framework.

System Development Methodology

The development of the system followed the classic Waterfall Software Development Life Cycle (SDLC) model. Each phase—commencing with requirements gathering, proceeding through architectural design, implementation, verification testing, and operational deployment—was executed systematically to ensure rigorous alignment with the operational requirements of Boss Rap Motor Shop.

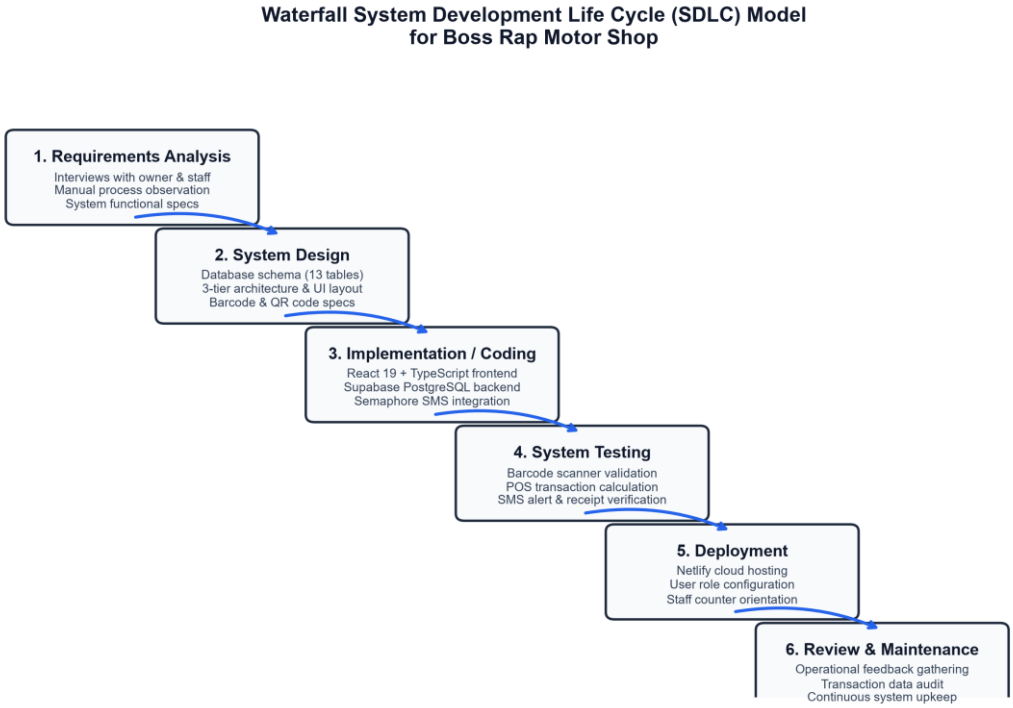


Figure 23: Waterfall SDLC Development Model for Boss Rap Motor Shop

Three-Tier System Architecture

The system is organized into a three-tier web architecture comprising the Presentation Layer, the Application and Business Logic Layer, and the Data and Cloud Storage Layer. The Presentation Layer encompasses the responsive user interfaces for administrators, counter staff, registered members, and logistics riders. The Application Layer executes transactional business rules, loyalty points calculation, barcode parsing, and SMS gateway integration. The Data Layer manages 13 relational tables in Supabase PostgreSQL alongside secure cloud bucket storage.

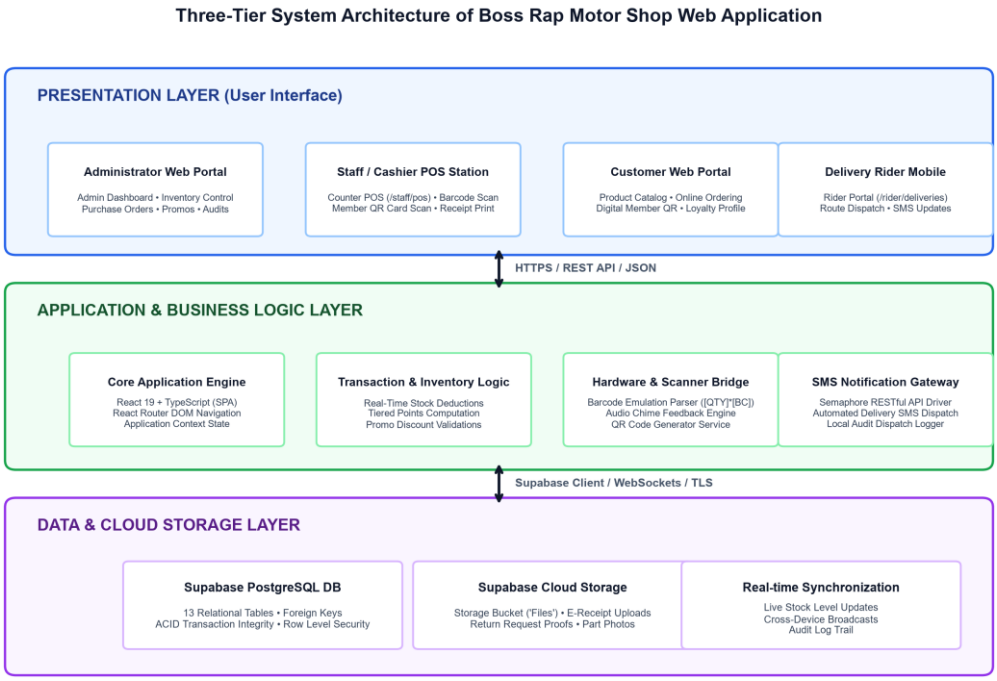


Figure 24: Three-Tier Web Architecture of Boss Rap Motor Shop System

Entity Relationship Diagram (ERD)

The database architecture is designed with strict relational integrity, normalized structures, and foreign key dependencies across 13 core tables. The central entities include `members`, `products`, `accounts`, `sales`, `sale_items`, `purchase_orders`, `purchase_order_items`, `return_requests`, `promos`, `points_settings`, `inquiries`, `audit_logs`, and `back_orders`.



Figure 25: Entity Relationship Diagram (ERD) of Boss Rap Motor Shop Database Schema

Database Table Specifications

Table Name	Primary Key	Foreign Keys	Functional Description
members	id (text)	None	Stores customer profile, contact, address, registration date, and cumulative loyalty points
products	id (text)	None	Maintains parts inventory catalog, categories, selling prices, stock levels, and unique EAN-13 barcodes
accounts	id (text)	member_id -> members(id)	Controls system login credentials, user names, active status, and access roles (ADMIN, EMPLOYEE, CUSTOMER, RIDER)
sales	id (text)	member_id -> members(id)	Records transactions, totals, discounts, sales channels, fulfillment types, order statuses, and payment reference numbers

sale_items	id (uuid)	sale_id -> sales(id), product_id -> products(id)	Line-item breakdown of purchased parts, item quantities, and historical unit prices
purchase_orders	id (text)	None	Tracks supplier restocking purchase orders, supplier names, total cost, and delivery receipt dates
purchase_order_items	id (uuid)	purchase_order_id -> purchase_orders(id)	Line-item details of parts ordered from suppliers and unit cost prices
return_requests	id (text)	sale_id -> sales(id), member_id -> members(id)	Manages customer product warranty claims, return/replacement reasons, statuses, and proof attachments
promos	id (text)	None	Stores active promotional discount codes, discount percentages, minimum spend thresholds, and validity dates
points_settings	id (int)	None	Configures store-wide loyalty parameters: currency expenditure per point and monetary redemption value per point
inquiries	id (text)	None	Stores public messages and customer contact inquiries submitted via the website contact form
audit_logs	id (text)	None	Immutable chronological audit trail recording administrative logins, stock modifications, and critical system events
back_orders	id (text)	product_id -> products(id)	Logs customer reservations for currently out-of-stock motorcycle parts and notification statuses

Use Case Diagram

The Use Case Diagram defines the interactions between the four primary system actors—Store Admin, Store Cashier, Customer, and Delivery Rider—and the functional capabilities provided within the system boundary.

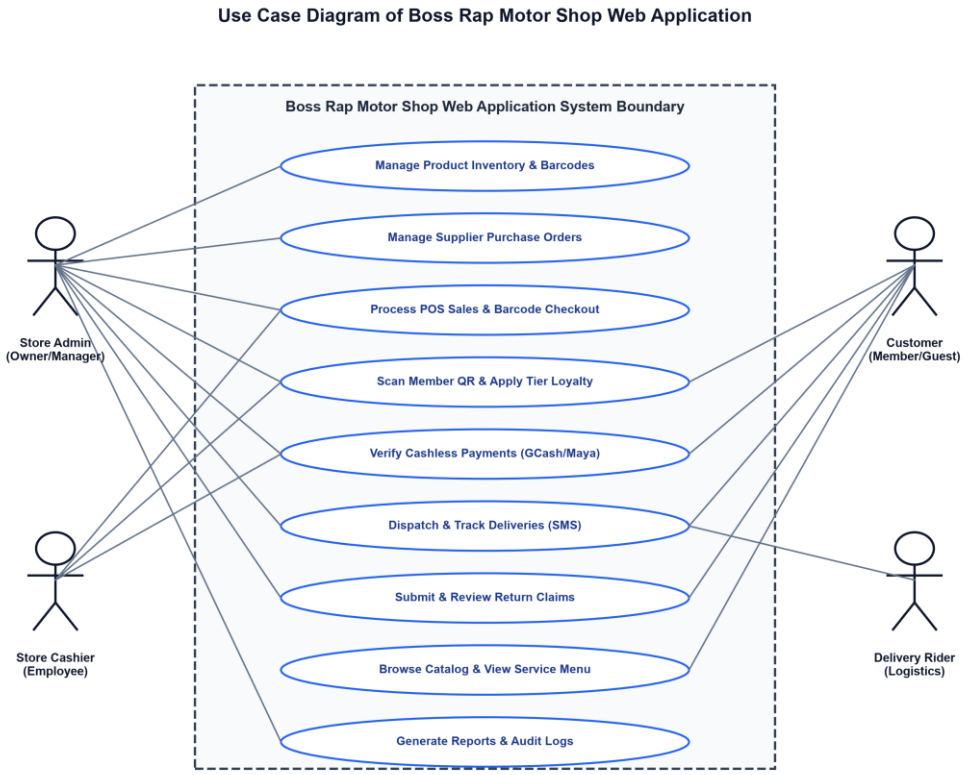


Figure 26: System Use Case Diagram of Boss Rap Motor Shop Web Application

Operational Process Flowcharts

To illustrate end-to-end operational execution, flowcharts were constructed for both in-store counter POS cashiering and online order fulfillment with rider delivery and SMS notifications.

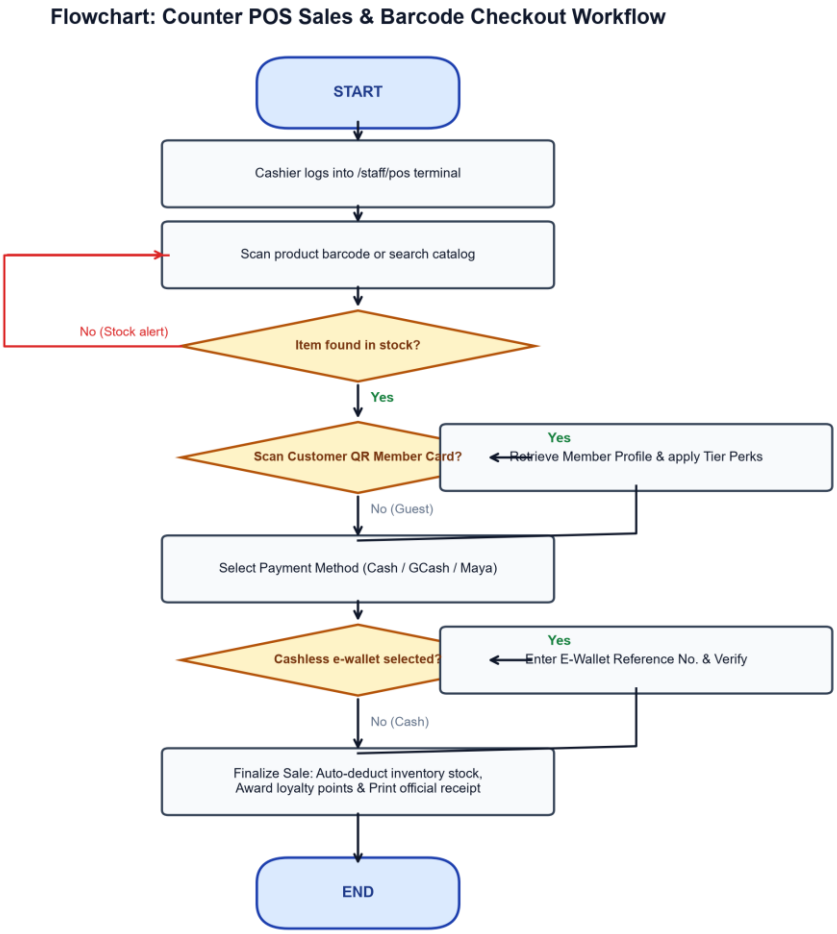


Figure 27: Flowchart of Counter POS Sales and Barcode Checkout Workflow

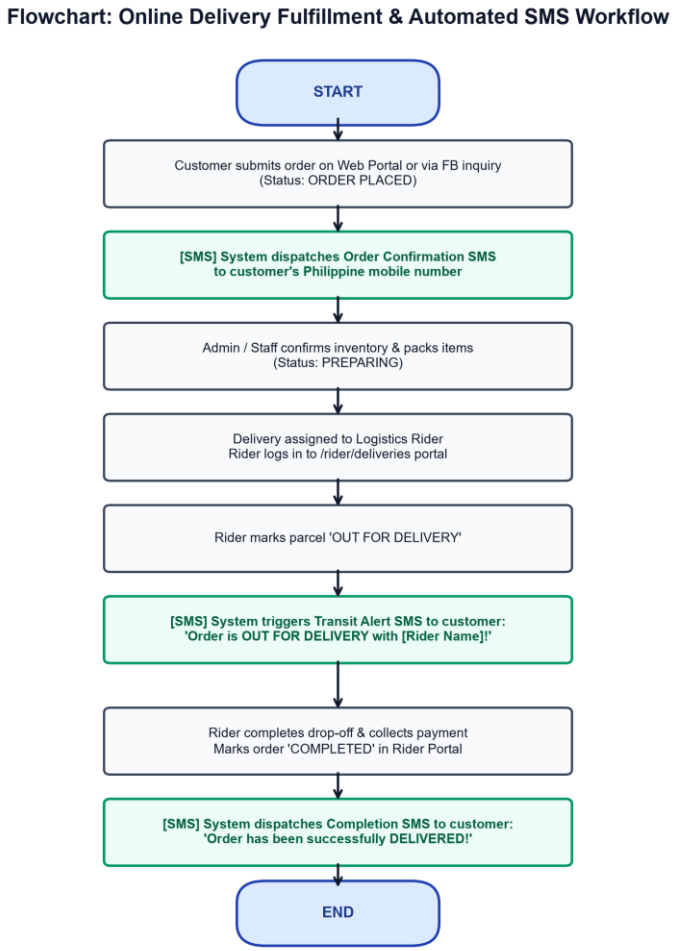


Figure 28: Flowchart of Online Delivery Fulfillment and Automated SMS Notification Workflow

Hierarchical Input-Process-Output (HIPO) Model

The HIPO model outlines the structural decomposition of the application, delineating the specific inputs, processing logic, and resulting outputs across each major functional module.

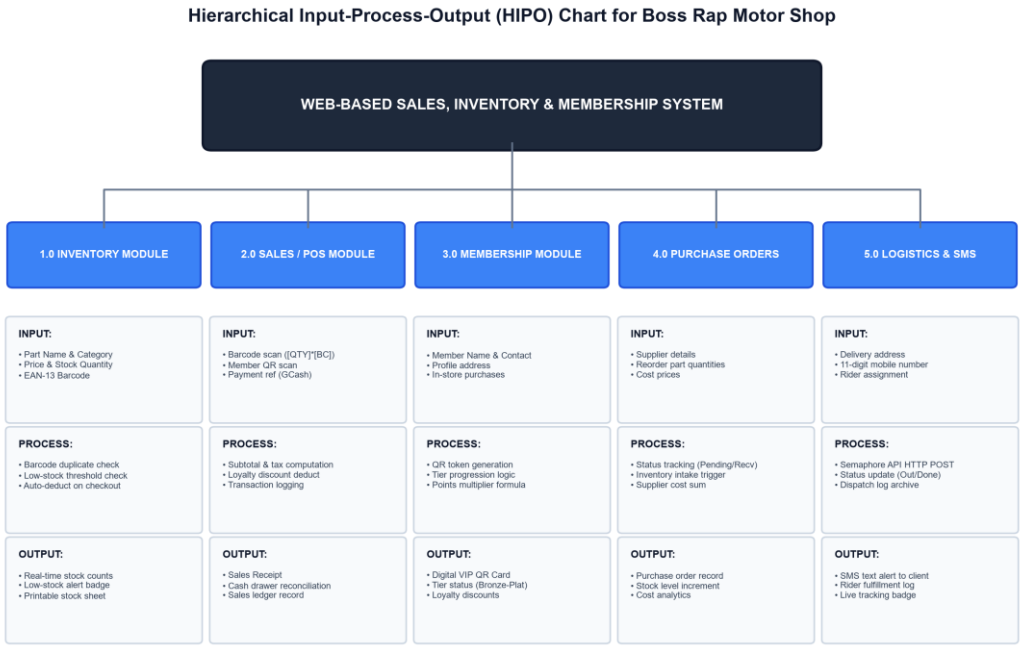


Figure 29: Hierarchical Input-Process-Output (HIPO) Model for Boss Rap Motor Shop

Survey Result

To evaluate the usability, functional suitability, payment reliability, and organizational impact of the Web-Based Sales and Inventory System with Membership Management, an empirical evaluation was conducted with the management, counter staff, and customers of Boss Rap Motor Shop. The survey instrument evaluated the system across four standard software quality dimensions using a 5-point Likert scale. Survey responses were gathered and analyzed using the weighted mean statistical formula.

System Field Testing, Demonstration, and Evaluation Documentation

To validate the operational viability, interface usability, and practical functionality of the Web-Based Sales and Inventory System with Membership Management, the proponents conducted three methodical on-site field visits, live application demonstrations, and empirical survey administrations at Boss Rap Motor Shop located in Baliuag, Bulacan. The first visit, conducted on October 14, 2026, focused on preliminary stakeholder consultations and operational assessment at the front service counter, examining manual stock logbooks, physical sales receipts, and counter workflows as illustrated in Figure 30. The second visit, conducted on October 18, 2026, involved an on-site live application demonstration and usability walkthrough on the shop's counter workstation (depicted in Figure 31), where store management and staff tested the real-time spare parts catalog, barcode checkout, and stock deduction features. The third visit, conducted on October 22, 2026, was dedicated to administering the standardized 20-

item evaluation questionnaire to motorcycle riders, mechanics, and patrons within the shop's service and waiting bays (shown in Figure 32). Comprehensive photographic exhibits and session logs are compiled in Appendix C.



Figure 30: Preliminary Stakeholder Consultation and Needs Assessment at Boss Rap Motor Shop Parts Counter (Conducted on October 14, 2026)



Figure 31: On-Site Live Web Application Demonstration and Usability Walkthrough on Counter Workstation (Conducted on October 18, 2026)



Figure 32: Administration of Survey Evaluation Questionnaires to Motor Shop Personnel, Riders, and Customers (Conducted on October 22, 2026)

Weighted Mean Formula

The evaluation responses were tabulated and computed using the standard weighted mean formula:

$$\bar{X}=\Sigma(f\cdot x)/N$$

Where: $\bar{X}$ = Computed Weighted Mean rating for each indicator

f = Frequency of responses assigned to a specific rating scale value

x = Numeric rating scale weight (5 = Strongly Agree, 4 = Agree, 3 = Moderately Agree, 2 = Disagree, 1 = Strongly Disagree)

N = Total number of participating survey respondents

Rating Scale and Descriptive Interpretation Table

Rating Scale Interval	Assigned Value (x)	Descriptive Verbal Interpretation
4.21 – 5.00	5	Strongly Agree / Excellent
3.41 – 4.20	4	Agree / Above Average
2.61 – 3.40	3	Moderately Agree / Average
1.81 – 2.60	2	Disagree / Below Average
1.00 – 1.80	1	Strongly Disagree / Poor

Usability and User Experience Evaluation

Table 5.1. Demographic Profile of Survey Respondents by Motor Shop Visit Frequency (N=102)

Frequency of Motor Shop Visits	Frequency (f)	Percentage (%)	Descriptive Interpretation
Often (Weekly / Regular Maintenance)	8	7.8%	Frequent Customer / Mechanic
Sometimes (Monthly / Periodic Services)	51	50.0%	Regular Customer / Motorist
Rarely (As-Needed Replacement)	38	37.3%	Occasional Customer
Never (First-Time Visitor)	5	4.9%	New Evaluator
Total	**102**	**100.0%**	**Total Valid Sample (N=102)**

As detailed in Table 5.1, the survey included a total of 102 respondents comprising motor shop

customers, regular riders, mechanics, and counter staff. A majority of 50.0% (f=51) visit or transact with the shop periodically ('Sometimes'), while 37.3% (f=38) purchase parts as needed ('Rarely'), 7.8% (f=8) transact frequently ('Often'), and 4.9% (f=5) are first-time evaluators ('Never'). Combined, 95.1% of the sample represent experienced motorcycle parts consumers, ensuring that feedback gathered on user interface, barcode POS checkout, and loyalty card incentives is grounded in practical operational requirements.

The Usability and User Experience category evaluates interface clarity, ease of learning for new staff, and responsive accessibility across desktop and mobile screens.

Item No.	Evaluation Criteria: Usability and User Experience	5	4	3	2	1	Overall Mean	Verbal Interpretation
1	Navigation Ease: The system was easy to navigate across all sections (Shop, Services, Dashboard).	78	18	5	0	1	4.69	Strongly Agree
2	Search Efficiency: I was able to find motorcycle parts, maintenance services, and product information quickly.	63	33	5	0	1	4.54	Strongly Agree
3	Guidance & Controls: Buttons, links, and navigation controls clearly guided me through browsing and placing orders.	84	11	5	0	2	4.72	Strongly Agree
4	Information Clarity: Product prices, stock levels, and service descriptions were clear and easy to understand.	61	29	9	1	2	4.43	Strongly Agree
5	Visual Design: The overall visual design of the system was attractive and appropriate for a motorcycle shop.	81	16	3	1	1	4.72	Strongly Agree
6	Readability: Layout, typography, colors, and interface elements made important product and order details easy to read.	71	24	6	0	1	4.61	Strongly Agree
	Overall Category Weighted Mean Rating						**4.62**	**Strongly Agree**

Functional Suitability and Process Efficiency Evaluation

The Functional Suitability dimension examines whether the system's core capabilities—including inventory tracking, barcode item lookup, and purchase ordering—effectively satisfy business needs.

Item	Evaluation Criteria: Functional	5	4	3	2	1	Overall	Verbal

No.	Suitability & Efficiency						Mean	Interpretation
1	Interface Consistency: Product catalogs and transaction details maintained structured, coherent visual hierarchy.	62	36	3	0	1	4.55	Strongly Agree
2	Multi-Device Responsiveness: The system displayed properly and remained responsive on desktop, tablet, and mobile devices.	71	24	5	0	2	4.59	Strongly Agree
3	System Stability & Speed: The platform loaded quickly and operated smoothly without technical glitches or unexpected errors.	69	22	8	0	3	4.51	Strongly Agree
	Overall Category Weighted Mean Rating						**4.55**	**Strongly Agree**

Payment Reliability and System Communication Evaluation

This domain measures the dependability of cashless payment verification, official receipt printing, customer QR membership card retrieval, and automated SMS alerts.

Item No.	Evaluation Criteria: Payment & Communication	5	4	3	2	1	Overall Mean	Verbal Interpretation
1	Order Processing: The overall process of processing or placing an order was smooth and efficient.	62	35	3	1	1	4.53	Strongly Agree
2	Scanning Acceleration: Barcode scanning for items and QR code scanning for members significantly improved checkout speed and accuracy.	68	27	4	0	3	4.54	Strongly Agree
3	Payment Flexibility: Available payment methods (Cash, E-Wallet / GCash, Online Banking) were convenient and easy to use.	68	26	5	1	2	4.54	Strongly Agree
4	Order Communication: Order tracking, sales receipts, and notifications regarding order status were clear and delivered promptly.	69	28	3	0	2	4.59	Strongly Agree
	Overall Category Weighted Mean Rating						**4.55**	**Strongly Agree**

System Adoption and Organizational Impact Evaluation

This category evaluates the readiness of Boss Rap Motor Shop to adopt the web

platform as a permanent operational solution and its impact on business productivity.

Item No.	Evaluation Criteria: System Adoption & Business Impact	5	4	3	2	1	Overall Mean	Verbal Interpretation
1	Membership Access: Registration for membership and accessing the digital QR code membership card was quick and hassle-free.	60	27	12	0	3	4.38	Strongly Agree
2	Loyalty Rewards Value: Points accumulation, loyalty reward tiers, and promotional discounts provided real value.	71	19	10	0	2	4.54	Strongly Agree
3	History Tracking: Membership features made it easier to track purchase history, points earned, and ongoing order updates.	63	32	5	0	2	4.51	Strongly Agree
4	Expectations Fulfillment: Overall, the system met my expectations in terms of usability, convenience, and efficiency for motor shop management.	70	27	4	0	1	4.62	Strongly Agree
5	System Recommendation: I would recommend the Boss Rap Motor Shop Sales, Inventory, and Membership System to other customers or business operators.	72	24	5	0	1	4.63	Strongly Agree
	Overall Category Weighted Mean Rating						**4.54**	**Strongly Agree**

Qualitative Feedback and User Improvement Suggestions

In Question 20 of the evaluation questionnaire, respondents provided qualitative insights and actionable suggestions for future system enhancement. The qualitative feedback was systematically categorized into recurring thematic areas to guide future developmental iterations:

Thematic Area	Key Respondent Suggestions (Question 20)	Proponent Action / Planned Roadmap
Inventory Automation & Alerts	Implement automatic low-stock notifications and alerts whenever fast-moving spare parts fall below threshold quantities.	Fully implemented in Supabase real-time triggers and low-stock dashboard status badges.
Hardware Barcode Scanning	Introduce dedicated laser barcode guns at the cash counter to further accelerate checkout during peak shop hours.	Included in Chapter 5 hardware recommendations for physical store upgrade.
Multi-Channel Digital Payments	Expand payment gateway integrations beyond manual reference logging to include automated QR Ph dynamic merchant checkout.	Targeted for Phase 2 implementation via PayMongo and Maya merchant APIs.
Mobile App Companion	Develop an Android companion app for	Documented under future

	motorcycle delivery riders and customers to track orders via push notifications.	recommendations using React Native or Flutter.

Overall Evaluation Summary Table

Category No.	Evaluation Domain	Composite Weighted Mean	Verbal Interpretation
1	Usability and User Experience	4.62	Strongly Agree / Excellent
2	System Performance and Technical Reliability	4.55	Strongly Agree / Excellent
3	Core Functional Suitability, POS & Payment Methods	4.55	Strongly Agree / Excellent
4	Membership, Loyalty, & System Recommendation	4.54	Strongly Agree / Excellent
	General Composite Weighted Mean Rating	**4.56**	**Strongly Agree / Highly Acceptable**

Summary

This capstone study was conducted to design, develop, and implement the Web-Based Sales and Inventory System with Membership Management for Boss Rap Motor Shop, a motorcycle parts and maintenance business situated in Baliuag, Bulacan. Prior to this implementation, the motor shop operated through traditional, manual methods relying on physical paper notebooks to log inventory stock levels and record daily sales. Orders received through the store's Facebook page were tracked informally, creating risks of stock miscounts, slow transaction times, unorganized sales records, and a total absence of structured customer profiling.

To resolve these operational challenges, the proponents developed an integrated web-based platform using React 19, TypeScript, and Vite for the frontend user interface, interfaced with a Supabase PostgreSQL relational cloud database. The system incorporates an automated Point of Sale (POS) cashiering terminal with barcode scanner emulation, real-time stock deduction, low-stock threshold monitoring, supplier purchase ordering, and cashless payment tracking (GCash, Maya, and Online Bank Transfer). Furthermore, the platform establishes a structured customer membership

system utilizing dynamic QR code cards with automated loyalty tier progression (Bronze, Silver, Gold, Platinum). A dedicated delivery logistics portal was created for delivery riders, fully synchronized with the Semaphore SMS Gateway to dispatch real-time order confirmation, out-for-delivery transit, and delivery completion text alerts to Philippine mobile numbers. The system's operational viability was affirmed through rigorous testing and stakeholder evaluation, demonstrating that computerization effectively elevates the operational efficiency, inventory precision, and customer service delivery of Boss Rap Motor Shop.

Conclusion

Based on the project's development and operational evaluation against its established objectives, the following conclusions were reached:

1. The comprehensive Inventory Management Module successfully achieved real-time monitoring of motorcycle spare parts and accessories. By automating stock deductions upon POS sales and stock increments upon purchase order intake, the system eradicated handwritten logbook errors and established accurate visibility over stock levels.
2. The Sales Management and POS Terminal Module significantly enhanced transaction speed and checkout accuracy. Integration of barcode scanning emulation eliminated manual price lookups, while multi-channel payment logging provided transparent accounting for cash and digital e-wallet transactions.
3. The Membership Management and Loyalty Module effectively introduced structured customer relationship management. Issuance of unique QR code member cards enabled instant customer identification at the counter, accurately tracked purchase history, and incentivized repeat visits through tiered membership discounts.
4. The Reporting Module successfully automated the generation of vital business intelligence reports. The shop owner can instantly review daily sales performance, inventory turnover, fast-moving and slow-moving items, and comprehensive transaction logs, supporting informed, data-driven decisions.
5. The integrated synchronization across inventory, sales, purchase orders, logistics, and SMS alerts established a unified operational workflow. Connecting counter cashiering with rider delivery dispatch and automated customer telecommunications eliminated the disconnect between in-store operations and online Facebook customer inquiries.

Recommendations

To maximize the long-term operational impact and sustainability of the Web-Based Sales and Inventory System with Membership Management, the proponents put forward the following recommendations:

1. Hardware Upgrades: The shop is encouraged to install dedicated, multi-directional USB/Bluetooth laser barcode scanners and heavy-duty 80mm thermal receipt printers at the counter to further accelerate checkout processing during peak shop hours.

2. Native Mobile Applications: Future developers may expand the web platform by building companion mobile apps (using React Native or Flutter) for delivery riders and customers, enabling GPS route navigation for riders and push notifications for registered members.

3. Integration with Direct Banking APIs and QR Ph: Expanding beyond manual reference number entry by integrating direct merchant payment gateways (such as PayMongo or Maya Business Checkout) will allow automated instant payment webhooks and automated status reconciliation.

4. Predictive Inventory Analytics: Future enhancements should incorporate machine learning or time-series sales forecasting algorithms to predict seasonal parts demand and automatically recommend purchase order quantities ahead of peak motorcycle riding seasons.

5. Staff Training & Database Maintenance: Regular database indexing, cloud backup schedules, and periodic staff retraining on cybersecurity best practices must be observed to ensure data integrity, system security, and continuous system reliability.

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Appendices

APPENDIX A. RESOURCE PERSONS

The following individuals served as resource persons, academic mentors, review panel members, and industry stakeholders whose guidance, technical critique, and institutional support contributed to the successful completion of this capstone project:

Name of Resource Person	Designation / Role	Institution / Organization
Mr. Jovie Mart S. Salas	Capstone Project Adviser	Information and Communications Technology Program, STI College Baliuag
Mr. Kar-el T. Paulino	Capstone Project Coordinator / Program Head	Information and Communications Technology Program, STI College Baliuag
Mr. Jayson V. Villafuerte	Capstone Review Panel Member	Information and Communications Technology Program, STI College Baliuag
Mr. Louie A. Ruiz	Lead Panelist, Capstone Review Panel	Information and Communications Technology Program, STI College Baliuag
Proprietor / Management	Store Owner & Client Stakeholder	Boss Rap Motor Shop, Baliuag, Bulacan
Store Personnel & Mechanics	Operational End-Users / Resource Evaluators	Boss Rap Motor Shop, Baliuag, Bulacan

APPENDIX B. PERSONAL TECHNICAL VITAE

This appendix contains the formal Curriculum Vitae of the four student researchers and system developers from the Bachelor of Science in Information Technology program at STI College Baliuag who designed, engineered, and evaluated the Web-Based Sales and Inventory System with Membership Management for Boss Rap Motor Shop:

Curriculum vitae of

Vladimmhir B. Canchino

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Canchino01@gmail.com
09458159270



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2022–Present	STI Baliuag College
Tertiary	2021–2022	Baliwag Polytechnic College
Vocational/Technical	2020–2022	Sulivan National High School
High School	2016–2019	Sulivan National High School
Elementary	2010–2016	Tilapayong Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

SKILLS

Skills	Level of Competency	Date Acquired
C#	Beginner	2024
HTML	Beginner	2026
CSS	Beginner	2024
SQL	Beginner	2024
JAVA	Beginner	2023

TRAININGS, SEMINARS, OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
February 2025	System Administration and Maintenance
September 2024	Network Technology

Curriculum vitae of

Aljae DJ. Cruz

Sta. Lucia Angat Bulacan
boychickc18@gmail.com
09750468893



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2022–Present	STI Baliuag College
Vocational/Technical	2020–2022	Lourdes College of Bulacan
High School	2017–2020	Lourdes College of School
Elementary	2011–2017	Benito C. Cruz Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

SKILLS

Skills	Level of Competency	Date Acquired
C#	Beginner	2024
HTML	Beginner	2026
CSS	Beginner	2024
SQL	Beginner	2024
JAVA	Beginner	2023

TRAININGS, SEMINARS, OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
February 2025	System Administration and Maintenance
September 2024	Network Technology

Curriculum vitae of

Mohammad Hamdie T. Langco

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Langcohamdie@gmail.com
09974988035



EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2022–Present	STI Baliuag College
Vocational/Technical	2020–2022	Marian College
High School	2016–2019	San Benilde Integrated School
Elementary	2010–2016	Virgen Delas Flores Elementary School

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

SKILLS

Skills	Level of Competency	Date Acquired
C#	Beginner	2024
HTML	Beginner	2026
CSS	Beginner	2024
SQL	Beginner	2024
JAVA	Beginner	2022

TRAININGS, SEMINARS, OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
February 2025	System Administration and Maintenance
September 2024	Network Technology

Curriculum vitae of

Don Janken B. Teodoro

Bonga Menor, Bustos, Bulacan
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EDUCATIONAL BACKGROUND

Level	Inclusive Dates	Name of School/Institution
Tertiary	2023–Present	STI College Baliuag
Vocational/Technical	2020–2021	Aguinaldo J. Santos National High School
Vocational/Technical	2019–2020	Integrated College - Baliwag
High School	2016–2019	Mary and Jesus School Inc.
Elementary	2015–2016	Aguinaldo J. Santos National High School
Elementary	2009–2015	Bonga Menor Elementary School
Elementary	2008–2009	Marren School of Bulacan

PROFESSIONAL OR VOLUNTEER EXPERIENCE

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

AFFILIATIONS

Inclusive Dates	Name of Organization	Position
N/A	N/A	N/A

SKILLS

Skills	Level of Competency	Date Acquired
C#	Beginner	2024
HTML	Beginner	2026
CSS	Beginner	2024
SQL	Beginner	2024
JAVA	Beginner	2023

TRAININGS, SEMINARS, OR WORKSHOP ATTENDED

Inclusive Dates	Title of Training, Seminar, or Workshop
February 2025	System Administration and Maintenance
September 2024	Network Technology

APPENDIX C. FIELD DOCUMENTATION AND PHOTO GALLERY

This appendix provides photographic documentation of the empirical research activities, stakeholder consultations, hands-on system demonstrations, and survey questionnaire administrations conducted by the proponents at Boss Rap Motor Shop in Baliuag, Bulacan. These exhibits provide visual verification of the authentic field engagement, the qualitative investigation into manual record-keeping bottlenecks, and the rigorous on-site validation of the Web-Based Sales and Inventory System with Membership Management across 102 respondents.

Plate C.1: Preliminary Stakeholder Consultation and Needs Assessment



Plate C.1. Proponents conferring with Boss Rap Motor Shop personnel at the service counter to analyze manual records and operational bottlenecks (Conducted on October 14, 2026).

Field / Metric	Details and Field Activity Notes
Activity Title	Preliminary Stakeholder Consultation, Needs Assessment, and Physical Records Examination
Date & Venue	October 14, 2026 | Boss Rap Motor Shop Main Counter & Spare Parts Store, Baliuag, Bulacan
Proponents Present	Vladimmhir B. Canchino, Aljae Dj. Cruz, Mohammad Hamdie T. Langco, Don Janken B. Teodoro
Stakeholders Involved	Proprietor / Store Manager and Front Counter Clerk, Boss Rap Motor Shop

Activity Description & Operational Findings	Proponents in official STI College uniform conducted an on-site consultation and observational study at the main service counter. The researchers examined physical logbooks used for stock recording, reviewed manual paper receipts, and observed the counter sales workflow and GCash payment verifications. The store manager highlighted frequent delays during peak hours caused by manual item lookup and emphasized the necessity of barcode scanning and automated low-stock warnings.

Plate C.2: On-Site Web System Demonstration and Usability Testing



Plate C.2. Proponents demonstrating the live Web-Based Sales and Inventory System on the shop counter workstation (Conducted on October 18, 2026).

Field / Metric	Details and Field Activity Notes
Activity Title	Live Web Application Demonstration, System Walkthrough, and Operator Usability Testing
Date & Venue	October 18, 2026 | Boss Rap Motor Shop Administrative Workstation / Back Office, Baliuag, Bulacan
Proponents Present	Vladimmhir B. Canchino, Aljae Dj. Cruz, Mohammad Hamdie T. Langco, Don Janken B. Teodoro
Stakeholders Involved	Shop Owner / Head Administrator and Front Counter Personnel
Activity Description & Operational Findings	The researchers deployed the Web-Based Sales and Inventory System on the shop's computer terminal and guided the client through the live Inventory Management & Spare Parts catalog on a wide curved display. Store management tested real-time stock deductions, automated reorder threshold alerts, POS transaction processing, and membership points tracking. The client validated system responsiveness, noting significant reductions in item lookup time and commending the clean layout.

Plate C.3: Field Survey Administration and Questionnaire Data Gathering



Plate C.3. Proponents administering the 20-item capstone evaluation survey questionnaires to motorcycle riders, mechanics, and shop patrons (Conducted on October 22, 2026).

Field / Metric	Details and Field Activity Notes
Activity Title	End-User Survey Questionnaire Administration and Empirical Acceptance Evaluation
Date & Venue	October 22, 2026 | Boss Rap Motor Shop Customer Waiting Bay and Service Bays, Baliuag, Bulacan
Proponents Present	Vladimmhir B. Canchino, Aljae Dj. Cruz, Mohammad Hamdie T. Langco, Don Janken B. Teodoro
Respondents Involved	N=102 Total Respondents (Motorcycle Owners, Delivery Riders, Mechanics, Counter Staff, and Shop Management)
Activity Description & Operational Findings	The researchers administered the standardized 20-item Likert-scale evaluation questionnaire to patrons, walk-in customers, delivery riders, and mechanics in the service bay. Proponents assisted respondents by clarifying evaluation criteria across system usability, functional suitability, payment reliability, and membership loyalty acceptability. The collected survey dataset (N=102) yielded an overall General Weighted Mean of 4.56, demonstrating 'Strongly Agree' and high operational acceptance.